EIGENFUNCTIONS, FREE BOUNDARIES, AND TIME-FREQUENCY LOCALIZATION

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Abstract. We study Gaussian time–frequency localization operators through their Bargmann–Fock representation and develop an inverse and free-boundary approach to their spectral theory.

We develop a perturbative inverse theory showing that functions sufficiently close to the ground-state Hermite function can be realized as eigenfunctions of localization operators. As a byproduct, we recover the classical disk characterization of Abreu and Dörfler in the simply connected setting by the same free-boundary method. As an application we prove that the stability estimate of Gomez-Guerra, Ramos and Tilli is sharp.

In the second part of the paper we apply these inverse results to Faber–Krahn-type problems for Gaussian time–frequency concentration. We prove that local maximizers of the first localization eigenvalue under a measure constraint are necessarily disks, and we establish existence of global maximizers via a concentration–compactness argument in the Bargmann–Fock space, recovering the Nicola–Tilli theorem from this perspective.

1. Introduction

The study of time–frequency concentration lies at the heart of modern harmonic analysis, with deep connections to time–frequency analysis, signal processing, mathematical physics, and phase–space methods; see for instance [21, 28, 17]. A central object in this theory is the *short-time Fourier transform* (STFT), defined for a function $f \in L^2(\mathbb{R})$ and a window $\varphi \in L^2(\mathbb{R})$ by

$$V_\varphi f(x,\omega) = \int_{\mathbb{R}} f(t)\,\varphi(x-t)\,e^{-2\pi i t\omega}\,dt.$$

This transform measures the joint time–frequency content of f, and its squared modulus $|V_\varphi f|^2$ provides a phase–space energy density. Understanding how this energy can be concentrated in regions of the time–frequency plane is a fundamental problem, with both theoretical and applied significance.

In applications ranging from signal recovery to uncertainty principles, one is often interested in maximizing the quantity

$$\int_\Omega |V_\varphi f(x,\omega)|^2\,dx\,d\omega$$

over functions f of unit norm and sets Ω of prescribed measure. This leads naturally to the study of *time–frequency localization operators*, introduced by Daubechies in [15] and further developed in [16, 33, 34, 14, 12, 7, 25, 32, 35]. For general windows, obtaining sharp concentration bounds is difficult, and Lieb's uncertainty principle [27] remains a basic benchmark; see also [2] for related large-sieve phenomena in modulation and polyanalytic Fock spaces. In the Gaussian setting, however, these operators enjoy a particularly rigid structure: via the Bargmann transform [4], they can be realized as integral operators on the Bargmann–Fock space, and the problem becomes

one of complex analysis in weighted spaces; see [8, 9, 10, 11, 41, 21, 17]. In particular, the STFT becomes intertwined with an entire function whose weighted modulus captures the concentration properties of the signal, allowing geometric questions to be translated into analytic ones.

A central theme in this area is the interplay between *spectral properties* of localization operators and the *geometry* of the underlying domains. In the Gaussian case, Nicola and Tilli proved the sharp Faber–Krahn inequality for the STFT, showing that disks maximize concentration among all sets of fixed measure [31]; see also [32, 5, 6] for complementary spectral and regularity results on localization operators. The stability theory for this variational problem was subsequently developed in [19], where quantitative control of both the first eigenfunction and the Fraenkel asymmetry is obtained from the spectral deficit. Related concentration questions for other transforms, including wavelet transforms, have been investigated in [36].

From a complementary perspective, one may ask inverse questions: to what extent does spectral information determine the geometry of the localization domain? Conversely, given a function, can one construct a domain for which it is an eigenfunction of the associated localization operator? The first systematic result in this direction is due to Abreu and Dörfler [1], who showed that if a Hermite function is an eigenfunction of a localization operator on a simply connected domain, then the domain must be a disk. This inverse viewpoint is closely related to weighted quadrature identities and free-boundary problems in potential theory; compare [3, 13, 22, 39, 38, 24]. It is also closely tied to the special role of Gaussian decay and Hermite structures in the Bargmann model; see [26, 20, 40].

The guiding principle of the present paper is that, for Gaussian localization operators, eigenfunctions should be regarded as geometric data. More precisely, the eigenfunction should determine the domain in rigid situations, and the possibility of prescribing the eigenfunction should in turn have strong consequences for nearby variational problems. The three main threads of the paper are organized around this idea:

(1) rigid eigenfunctions force rigid domains;
(2) near the Gaussian, one can prescribe the eigenfunction and construct the corresponding localization domain;
(3) once such an eigenfunction-to-domain mechanism is available, it can be used to analyze sharpness and extremizers in concentration inequalities.

The present work contributes to this line of research by developing a unified inverse and free-boundary framework for Gaussian time–frequency localization operators. Our approach builds on and extends the rigidity result of Abreu and Dörfler [1], the variational theory of Nicola and Tilli [31, 32], and the quantitative stability results of Gómez, Guerra, Ramos, and Tilli [19]. At the technical level, our arguments also draw on ideas from inverse potential problems and quadrature-domain theory, in the spirit of [23, 13, 22, 39, 38]. In this way, the paper places localization operators squarely within the broader free-boundary and inverse-potential framework suggested by the Gaussian Bargmann model.

Despite these advances, several fundamental questions have remained open. In particular, the construction of localization domains associated with prescribed functions and the role of

eigenfunction rigidity in variational concentration problems have not been fully understood. The results of the present paper aim to address these questions in a unified manner.

Our first main result is the constructive half of this picture. It shows that, near the Gaussian, one can indeed prescribe the eigenfunction and realize it by a suitable localization domain.

Theorem 1. *Let $\lambda \in (0,1)$ and $N \in \mathbb{N}$. There exists $\varepsilon_0(\lambda, N) > 0$ such that if*

$$f_0 = h_0 + \sum_{k=1}^{N} a_k h_k$$

satisfies $\max_k |a_k| < \varepsilon_0$, then there exists a domain U_λ such that

$$\mathcal{L}_{U_\lambda} f_0 = \lambda f_0.$$

This theorem provides, to the best of our knowledge, the first systematic construction of localization domains associated with nontrivial linear combinations of Hermite functions. In particular, it supplies the inverse counterpart to the rigidity phenomenon of Abreu and Dörfler: rather than deducing the domain from a rigid eigenfunction, we construct domains from prescribed near-Gaussian eigenfunctions.

The proof proceeds in several stages, combining ideas from free-boundary problems and complex analysis. First, we reformulate the eigenvalue equation in the Bargmann–Fock space as a nonlinear integral identity involving the squared modulus of a polynomial. By applying a suitable differential operator, we reduce the problem to constructing a domain satisfying a weighted identity reminiscent of a quadrature condition.

We then construct an approximate solution by solving a free-boundary problem of obstacle type. This is achieved via a perturbative theorem of Isakov, which guarantees the existence of a domain whose associated potential satisfies a prescribed equation under small perturbations of the data. This step produces a domain for which the desired identity holds up to a controlled error.

In a second step, we refine this construction using a weighted quadrature-domain theorem. By viewing the problem through the lens of complex analysis, we interpret the identity as a condition on a weighted Cauchy transform. A perturbative implicit-function argument then allows us to deform the domain so that the identity holds exactly. Finally, we show that the resulting function satisfies the original eigenvalue equation by analyzing an associated differential operator and ruling out nontrivial solutions via growth estimates. The combination of these steps yields a robust local inverse theory.

A noteworthy aspect of this construction is that it allows us to go beyond monomial-type eigenfunctions. Indeed, previous constructions in the literature were largely restricted to highly symmetric situations. Our result shows that a much broader class of functions can be realized spectrally, opening the door to a richer inverse theory and suggesting new connections with nonlinear potential theory.

The same free-boundary framework also yields the rigid half of the correspondence, namely a short recovery of the classical theorem of Abreu and Dörfler.

More precisely, we provide two new proofs of the Abreu–Dörfler rigidity result based on our methods: one through the local radiality mechanism for the logarithmic potential, and a second through a holomorphic auxiliary function attached to the associated Poisson problem.

Theorem 2. *Let U be a simply connected bounded domain. If a Hermite function is an eigenfunction of the localization operator associated with U, then U is a disk.*

The proof proceeds by transporting the eigenvalue equation to the Bargmann–Fock space, where Hermite functions correspond to monomials and the localization operator becomes an integral operator with reproducing kernel structure. In this setting, the eigenvalue equation can be interpreted as a functional identity involving exponential generating functions. By differentiating appropriately and invoking a Paley–Wiener type argument, we reduce the problem to the vanishing of a Fourier transform along complex lines.

This allows us to apply a divisibility lemma of Ehrenpreis–Malgrange type, which yields the existence of a logarithmic potential solving a Poisson equation with source supported on the domain. The key insight is that the spectral assumption imposes strong constraints on this potential: it must exhibit a form of radial behavior on the boundary. In the simply connected setting, this forces the domain to be a disk, thereby recovering the Abreu–Dörfler rigidity theorem from the same inverse/free-boundary perspective.

We now turn to consequences of this inverse point of view. Our first application concerns the quantitative stability theorem of Gómez, Guerra, Ramos, and Tilli for the Faber–Krahn inequality for the Gaussian STFT. In the formulation most relevant here, if $U \subset \mathbb{R}^2$ is a measurable set of finite positive measure, $\lambda_1(U)$ denotes the first eigenvalue of the associated localization operator, and f_U is a corresponding L^2-normalized first eigenfunction, then Corollary 1.4 of [19] gives the bounds

$$\min_{z_0\in\mathbb{C},\ |c|=1} \|f_U - c\phi_{z_0}\|_{L^2(\mathbb{R})} \le Ce^{|U|/2}\left(1-\frac{\lambda_1(U)}{1-e^{-|U|}}\right)^{1/2},$$

and

$$\mathcal{A}(U) \le K(|U|)\left(1-\frac{\lambda_1(U)}{1-e^{-|U|}}\right)^{1/2},$$

where ϕ_{z_0} ranges over the Gaussian optimizers and $\mathcal{A}(U)$ denotes the Fraenkel asymmetry of U.

Our perturbative inverse theorem provides the missing input needed to test this stability estimate on genuine spectral data. Starting from a prescribed perturbation of the Gaussian, we construct an exact eigenpair $(U_\varepsilon, f_\varepsilon)$ for the localization operator, with U_ε a smooth perturbation of the disk and f_ε a corresponding first eigenfunction. Thus the constructive direction of the paper produces explicit near-extremizers, and allows us to compare the size of the spectral deficit with the actual geometric displacement of the domain from the class of disks. The resulting family shows that the square-root dependence in Corollary 1.4 is the best possible one.

Theorem 3. *The exponent $1/2$ in Corollary 1.4 of [19] is optimal. More precisely, the powers of the deficit*

$$1-\frac{\lambda_1(U)}{1-e^{-|U|}}$$

appearing in both the eigenfunction estimate and the Fraenkel-asymmetry estimate there cannot be replaced by β for any $\beta > 1/2$.

The argument is based on constructing a carefully chosen family of perturbations of the disk using the inverse theorem above. Starting from a polynomial perturbation of the ground-state Hermite function, we obtain a corresponding family of domains whose geometry deviates from that of a disk at first order. We then perform a detailed asymptotic analysis of the eigenvalue equation, computing its first variation via a transport formula and identifying the leading Fourier modes of the boundary deformation.

This analysis shows that the deformation is governed by second-order harmonics, which cannot be eliminated by translations. By comparing the perturbed domains with arbitrary disks, we obtain lower bounds on their symmetric difference that match the upper bounds in the stability estimate. The proof combines spectral perturbation theory, geometric measure estimates, and harmonic analysis on the circle, and illustrates how inverse constructions can be used to probe the sharpness of variational inequalities.

We next connect this eigenfunction-to-domain philosophy to variational problems. Our fourth main result concerns the structure of local maximizers for Gaussian time–frequency concentration.

Theorem 4. *Any local maximizer of $\lambda_{1,\varphi}$ under a measure constraint is, up to translation, a disk.*

The proof relies on combining variational arguments with the inverse spectral perspective developed earlier. We first show that any local maximizer must correspond to a superlevel set of the density associated with its first eigenfunction in the Bargmann–Fock space. This reduces the geometric problem to a functional one.

We then analyze the spectral properties of the localization operator at a maximizer, showing that the first eigenvalue is simple and that derivatives of the eigenfunction remain eigenfunctions. This strong rigidity property forces the maximizing eigenfunction into a Gaussian form. Once this is known, the corresponding domain is determined as well, and one concludes that the only admissible configuration is a disk. The argument thus shows that extremizers are governed by the same rigidity mechanism that underlies the inverse theory.

Finally, we address the global variational problem and recover the Nicola–Tilli theorem from the same perspective.

Theorem 5 (New proof of Nicola–Tilli)**.** *Disks maximize Gaussian time–frequency concentration among all sets of fixed measure.*

Our proof is based on a concentration–compactness argument in the Bargmann–Fock space. We consider maximizing sequences and analyze their possible limiting behavior, ruling out vanishing and dichotomy scenarios using the structure of the Fock space and the invariance properties of the STFT. This yields the existence of a maximizer. Related concentration–compactness methods have recently been used in nearby time–frequency concentration problems, notably by Nicola, Romero, and Trapasso for ambiguity-function type concentration functionals [30]; see

also [29, 37] for recent work on Fourier concentration operators, and [18] for closely related developments in the polyanalytic Fock-space setting.

We then combine this existence result with the rigidity of local maximizers established earlier. Since any global maximizer is, in particular, a local maximizer, its eigenfunction is rigid and therefore determines the maximizing domain. This provides a new conceptual route to the Nicola–Tilli theorem, avoiding rearrangement inequalities and instead relying on inverse and free-boundary methods. The argument highlights the power of treating eigenfunctions as geometric data.

The paper is organized as follows. In Section 2 we review the Bargmann–Fock representation and establish several auxiliary results, including a key divisibility lemma and perturbative free-boundary theorems. Section 3 is devoted to the perturbative inverse construction. Section 4 explains how the same machinery recovers the classical theorem of Abreu and Dörfler in the simply connected case. Section 5 contains the sharpness result, while the remaining sections are devoted to variational applications, culminating in the new proof of the Nicola–Tilli theorem.

In summary, the results of this paper establish a bridge between time–frequency analysis, inverse spectral theory, and free-boundary problems in potential theory. This perspective not only resolves several open problems, but also provides a flexible and robust framework for studying a wide range of questions in phase–space analysis. :contentReference[oaicite:0]index=0

2. Preliminaries

2.1. **Properties of the Bargmann-Fock space.** We briefly recall the connection between the Gaussian short-time Fourier transform (STFT) and the Bargmann–Fock space, which will be used systematically in what follows.

Let $\phi(x) = 2^{1/4}e^{-\pi x^2}$ be the normalized Gaussian. For $f \in L^2(\mathbb{R})$, its Gaussian STFT is

$$V_\phi f(x,\omega) = \int_{\mathbb{R}} f(t)\,\phi(x-t)\,e^{-2\pi i t\omega}\,dt. \tag{2.1}$$

The Bargmann transform $\mathcal{B} : L^2(\mathbb{R}) \to \mathcal{F}^2(\mathbb{C})$ is defined by

$$(\mathcal{B}f)(z) = 2^{1/4}\int_{\mathbb{R}} f(t)\,e^{2\pi t z - \pi t^2 - \frac{\pi}{2}z^2}\,dt, \qquad z \in \mathbb{C}. \tag{2.2}$$

It is well known that $\mathcal{B}$ is a unitary isomorphism onto the Bargmann–Fock space $\mathcal{F}^2(\mathbb{C})$, endowed with the norm

$$\|F\|^2_{\mathcal{F}^2} = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\,dA(z).$$

A direct computation shows that, for $z = x + i\omega$,

$$V_\phi f(x,-\omega) = e^{\pi i x\omega}\,(\mathcal{B}f)(z)\,e^{-\frac{\pi}{2}|z|^2}. \tag{2.3}$$

In particular, setting $F = \mathcal{B}f$, one obtains

$$\int_{\mathbb{R}^2} |V_\phi f(x,\omega)|^2\,dx\,d\omega = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\,dA(z) = \|F\|^2_{\mathcal{F}^2}. \tag{2.4}$$

Weyl operators and invariance. For $a \in \mathbb{C}$, define the operator

$$(T_a F)(z) := e^{\pi a z - \frac{\pi}{2}|a|^2} F(z-a). \tag{2.5}$$

Then T_a is unitary on $\mathcal{F}^2(\mathbb{C})$. Indeed,

$$\begin{aligned}\|T_a F\|_{\mathcal{F}^2}^2 &= \int_{\mathbb{C}} e^{2\pi\Re(az)-\pi|a|^2}\,|F(z-a)|^2 e^{-\pi|z|^2}\,dA(z)\\ &= \int_{\mathbb{C}} |F(z-a)|^2 e^{-\pi|z-a|^2}\,dA(z) = \|F\|_{\mathcal{F}^2}^2,\end{aligned}$$

because

$$2\Re(az) - |a|^2 - |z|^2 = -|z-a|^2.$$

Moreover,

$$u_{T_a F}(z) = u_F(z-a). \tag{2.6}$$

Indeed,

$$u_{T_a F}(z) = |T_a F(z)|^2 e^{-\pi|z|^2} = e^{2\pi\Re(az)-\pi|a|^2}|F(z-a)|^2 e^{-\pi|z|^2} = |F(z-a)|^2 e^{-\pi|z-a|^2}.$$

In particular, the functionals I_F defined by, for $s > 0$,

$$I_F(s) := \sup_{|\Omega|=s} \int_\Omega u_F\,dA,$$

satisfy

$$I_{T_a F}(s) = I_F(s), \qquad a \in \mathbb{C},$$

reflecting the invariance of the Gaussian STFT under time–frequency shifts.

Connection with localization. If $F = \mathcal{B}f$, then for every measurable $\Omega \subset \mathbb{C}$,

$$\int_\Omega u_F\,dA = \int_{\widetilde{\Omega}} |V_\phi f(x,\omega)|^2\,dx\,d\omega,$$

where $\widetilde{\Omega} = \{(x,\omega) : (x,-\omega) \in \Omega\}$. Thus, concentration properties of the STFT are naturally encoded by the function

$$u_F(z) := |F(z)|^2 e^{-\pi|z|^2}, \qquad z \in \mathbb{C}, \tag{2.7}$$

which is the main object of interest in the Bargmann–Fock setting, in terms of the quantities $I_F(s)$.

We next collect a few elementary facts that will be used repeatedly.

Lemma 2.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) The space $\mathcal{F}^2(\mathbb{C})$ is a reproducing kernel Hilbert space with kernel

$$K_w(z) = e^{\pi z \overline{w}}, \qquad z, w \in \mathbb{C}.$$

(ii) (Sharp pointwise bound) For every $z \in \mathbb{C}$,

$$u_F(z) \le \|F\|_{\mathcal{F}^2}^2,$$

and equality holds at some point if and only if F is proportional to F_w for some $w \in \mathbb{C}$.

(iii) $u_F(z) \to 0$ as $|z| \to \infty$.

(iv) One has the estimate

$$\|u_F - u_G\|_{L^1(\mathbb{C})} \leq \big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big)\|F - G\|_{\mathcal{F}^2}.$$

Proof. (i) For $w \in \mathbb{C}$, the function $K_w(z) = e^{\pi z\overline{w}}$ belongs to $\mathcal{F}^2(\mathbb{C})$, and

$$\|K_w\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} e^{2\pi\Re(z\overline{w})-\pi|z|^2}\, dA(z) = e^{\pi|w|^2},$$

after completing the square. Thus point evaluation is continuous, and for every $F \in \mathcal{F}^2(\mathbb{C})$ one has the reproducing identity

$$F(w) = \langle F, K_w\rangle_{\mathcal{F}^2}.$$

In particular F is entire.

(ii) is a consequence of the reproducing kernel identity and Cauchy–Schwarz, as in (**??**).

(iii) Fix $\varepsilon > 0$, and choose a polynomial P such that

$$\|F - P\|_{\mathcal{F}^2} < \varepsilon$$

using the density of polynomials in $\mathcal{F}^2(\mathbb{C})$. By the pointwise bound from (ii),

$$|F(z) - P(z)|e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon \qquad \text{for every } z \in \mathbb{C}.$$

Since P is a polynomial,

$$P(z)e^{-\frac{\pi}{2}|z|^2} \to 0 \qquad \text{as } |z| \to \infty.$$

Hence

$$\limsup_{|z|\to\infty} |F(z)|e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon.$$

As $\varepsilon > 0$ is arbitrary, we conclude that

$$|F(z)|e^{-\frac{\pi}{2}|z|^2} \to 0,$$

that is, $u_F(z) \to 0$ as $|z| \to \infty$.

(iv) Writing

$$|u_F - u_G| = \big||F|^2 - |G|^2\big|e^{-\pi|z|^2} \leq |F - G|(|F| + |G|)e^{-\pi|z|^2},$$

the claim follows by Cauchy–Schwarz. □

We now note the following crucial fact about the Bargmann transform, which is that it transforms the basis of normalized Hermite functions onto normalized monomials, yielding thus a natural orthonormal basis on $\mathcal{F}^2(\mathbb{C})$.

Lemma 2.2. *The functions*

$$e_k(z) := b_k z^k, \qquad k \geq 0,$$

form an orthonormal basis of $\mathcal{F}^2(\mathbb{C})$, where

$$b_k = \left(\frac{\pi^k}{k!}\right)^{1/2}.$$

In particular,

$$\langle e_k, e_m\rangle_{\mathcal{F}^2} = \delta_{k,m}.$$

Proof. A direct computation in polar coordinates gives

$$\int_{\mathbb{C}} |z|^{2k} e^{-\pi|z|^2} dA(z) = 2\pi \int_0^\infty r^{2k+1} e^{-\pi r^2} dr = \frac{k!}{\pi^k}.$$

Thus $\|z^k\|_{\mathcal{F}^2}^2 = \frac{k!}{\pi^k}$, which yields the normalization. Orthogonality follows from rotational invariance. Completeness is standard. □

Lemma 2.3. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) F admits the expansion

$$F(z) = \sum_{k=0}^{\infty} a_k e_k(z), \qquad \sum_{k=0}^{\infty} |a_k|^2 = \|F\|_{\mathcal{F}^2}^2.$$

(ii) (Growth bound) For every $z \in \mathbb{C}$,

$$|F(z)| \le \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) (Derivative bounds) For every $z \in \mathbb{C}$,

$$|F'(z)| \le C(1+|z|)\, \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

Proof. (i) follows from the orthonormal basis in Lemma 2.2.

(ii) follows from the reproducing kernel estimate:

$$|F(z)| = |\langle F, K_z\rangle| \le \|F\|_{\mathcal{F}^2} \|K_z\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) Since

$$F'(z) = \pi \langle F, \overline{w} K_z(w)\rangle_{\mathcal{F}^2},$$

Cauchy–Schwarz gives

$$|F'(z)| \le \pi \|F\|_{\mathcal{F}^2}\, \|\overline{w} K_z\|_{\mathcal{F}^2}.$$

Now

$$\|\overline{w} K_z\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} |w|^2 e^{2\pi \Re(w\overline{z}) - \pi|w|^2}\, dA(w) = e^{\pi|z|^2} \int_{\mathbb{C}} |w+z|^2 e^{-\pi|w|^2}\, dA(w),$$

after the change of variables $w \mapsto w+z$. The last integral is bounded by $C(1+|z|^2)$, so

$$|F'(z)| \le C(1+|z|)\, \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

□

Lemma 2.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$ and set*

$$T := \sup_{z \in \mathbb{C}} u_F(z).$$

Then $0 \le T \le \|F\|_{\mathcal{F}^2}^2$, and $T = \|F\|_{\mathcal{F}^2}^2$ if and only if F is proportional to F_w for some $w \in \mathbb{C}$.

Proof. The bound follows from Lemma 2.1(ii). The characterization of equality is the same as in the reproducing kernel inequality. □

We proceed with a few elementary facts.

Lemma 2.5. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) $I_F(s)$ is attained by a superlevel set of u_F, and

$$0 \le I_F(s) \le \|F\|_{\mathcal{F}^2}^2.$$

(ii) For every $c \in \mathbb{C}$,

$$I_{cF}(s) = |c|^2 I_F(s).$$

(iii) One has the Lipschitz bound

$$|I_F(s) - I_G(s)| \le \|u_F - u_G\|_{L^1(\mathbb{C})} \le \big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big)\, \|F - G\|_{\mathcal{F}^2}.$$

Proof. The identity

$$\int_{\mathbb{C}} u_F \, dA = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2} \, dA(z) = \|F\|_{\mathcal{F}^2}^2$$

is immediate from the definition of the Fock norm. Since $F \in \mathcal{F}^2$, the standard reproducing-kernel estimate implies that u_F is bounded and continuous, and in fact $u_F(z) \to 0$ as $|z| \to \infty$; thus $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$.

The existence of a maximizing set for $I_F(s)$, given by a superlevel set of u_F, is the classical bathtub principle. The bound

$$0 \le I_F(s) \le \int_{\mathbb{C}} u_F \, dA = \|F\|_{\mathcal{F}^2}^2$$

is immediate. The scaling identity in (ii) is obvious.

For (iii), for every measurable $\Omega \subset \mathbb{C}$ with $|\Omega| = s$,

$$\left| \int_\Omega u_F \, dA - \int_\Omega u_G \, dA \right| \le \int_{\mathbb{C}} |u_F - u_G| \, dA.$$

Taking suprema over $|\Omega| = s$ gives

$$|I_F(s) - I_G(s)| \le \|u_F - u_G\|_{L^1(\mathbb{C})}.$$

Finally,

$$\begin{aligned} \|u_F - u_G\|_{L^1(\mathbb{C})} &= \int_{\mathbb{C}} \big| |F|^2 - |G|^2 \big| \, e^{-\pi|z|^2} \, dA \\ &\le \int_{\mathbb{C}} |F - G|(|F| + |G|) e^{-\pi|z|^2} \, dA \\ &\le \|F - G\|_{\mathcal{F}^2} \big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big) \end{aligned}$$

by Cauchy–Schwarz. □

2.2. **From time-frequency analysis to free boundary problems.** We now use the following divisibility lemma of Ehrenpreis–Malgrange flavour.

In order to state it, we say that a compactly supported distribution ω is a *singular-continuous free measure* if it defines a Radon measure on $\mathbb{R}^2$ whose Lebesgue decomposition has only absolutely continuous and pure point parts, and whose pure point part consists of finitely many atoms.

Lemma 2.6. *Let $\omega \in \mathcal{S}'(\mathbb{R}^2)$ be a compactly supported singular-continuous free distribution such that*

$$\widehat{\omega}(w, \pm iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Then there exist a finite set $P \subset \mathbb{R}^2$ and a compactly supported function $u \in L^2(\mathbb{R}^2)$ such that

$$\Delta u = \omega$$

in the sense of distributions, and

$$u \in C(\mathbb{R}^2 \setminus P).$$

Proof. By the Paley–Wiener theorem, $\widehat{\omega}(\xi, \eta)$ is an entire function on $\mathbb{C}^2$ of exponential type. Hence it admits a power series expansion

$$\widehat{\omega}(\xi, \eta) = \sum_{k,l \geq 0} a_{k,l}\, \xi^k \eta^l.$$

Using the vanishing hypotheses on the complex lines $\eta = \pm i\xi$, we get

$$0 = \widehat{\omega}(w, \pm iw) = \sum_{n \geq 0} \left(\sum_{k=0}^{n} a_{k,n-k} (\pm i)^{n-k} \right) w^n,$$

and therefore, by comparing coefficients,

$$\sum_{k=0}^{n} a_{k,n-k} (\pm i)^{n-k} = 0, \qquad \forall n \geq 0.$$

For each $n \geq 0$, define

$$p_n(t) := \sum_{k=0}^{n} a_{k,n-k} t^k.$$

Then $p_n(\pm i) = 0$, so $p_n(t)$ is divisible by $t^2 + 1$. Thus we may write

$$p_n(t) = (t^2 + 1) q_n(t),$$

where q_n is a polynomial of degree at most $n - 2$. Reassembling the homogeneous pieces, it follows that

$$\widehat{\omega}(\xi, \eta) = (\xi^2 + \eta^2) F(\xi, \eta)$$

for some entire function F on $\mathbb{C}^2$.

To justify the reassembly near the origin, write

$$\widehat{\omega}(\xi, \eta) = \sum_{n \geq 0} H_n(\xi, \eta),$$

where H_n is the homogeneous polynomial of degree n,

$$H_n(\xi, \eta) := \sum_{k=0}^{n} a_{k,n-k}\, \xi^k \eta^{n-k}.$$

The argument above shows that each H_n vanishes on the two lines $\eta = \pm i\xi$, hence is divisible by $\xi^2 + \eta^2$. Thus

$$H_n(\xi, \eta) = (\xi^2 + \eta^2) G_{n-2}(\xi, \eta)$$

for some homogeneous polynomial G_{n-2} of degree $n-2$ (with $G_{n-2} \equiv 0$ for $n < 2$). Summing these homogeneous pieces gives an entire function

$$F(\xi,\eta) := \sum_{n\geq 2} G_{n-2}(\xi,\eta)$$

and the claimed factorization follows term by term.

Again by Paley–Wiener, there exist constants $C, c > 0$ such that

$$|\widehat{\omega}(\xi,\eta)| \leq Ce^{c(|\Im\xi|+|\Im\eta|)}.$$

Moreover, since $\widehat{\omega}$ has a zero of order at least two at the origin, the quotient F is entire there, and away from a fixed neighborhood of the origin we may divide by $\xi^2+\eta^2$. Therefore

$$|F(\xi,\eta)| \leq C' \frac{e^{c(|\Im\xi|+|\Im\eta|)}}{1+|\xi|^2+|\eta|^2},$$

after enlarging C' if necessary to absorb the bounded region near the origin. In particular, restricting to $(\xi,\eta) \in \mathbb{R}^2$, we obtain

$$|F(\xi,\eta)| \leq \frac{C''}{1+|\xi|^2+|\eta|^2},$$

and therefore $F|_{\mathbb{R}^2} \in L^2(\mathbb{R}^2)$.

Let $u \in L^2(\mathbb{R}^2)$ be such that $\widehat{u} = F$ on $\mathbb{R}^2$. Then, up to the harmless Fourier-transform normalisation constant,

$$\widehat{\Delta u}(\xi,\eta) = -(\xi^2+\eta^2)\widehat{u}(\xi,\eta) = -(\xi^2+\eta^2)F(\xi,\eta),$$

so after rescaling u by a nonzero real constant if necessary, we may assume that

$$\Delta u = \omega$$

in the sense of distributions.

Since ω is a compactly supported Radon measure, u agrees (up to an additive harmonic function, which must vanish by compact support) with the logarithmic potential of ω:

$$u(x,y) = c\, G * \omega(x,y),$$

where $G(z) = \log|z|$ is the Green kernel in $\mathbb{R}^2$ and $c \in \mathbb{R}$. The absolutely continuous part of ω produces a continuous logarithmic potential, and the only possible singularities come from the finitely many atoms in the pure point part. Hence

$$u \in C(\mathbb{R}^2 \setminus P),$$

where P is the finite support of the atomic part of ω. □

2.3. **Two perturbative auxiliary results.** In this appendix we prove two perturbative results used in the proof of Theorem **??**. The first is a free-boundary stability statement around a strictly negative obstacle-type solution. The second is a local realization theorem for weighted quadrature domains near a reference domain. Both statements are tailored to the setting of the present paper.

2.3.1. *A perturbative free-boundary theorem.* We begin with the perturbative free-boundary statement used below. The domain is chosen so that the exterior Cauchy transform of the source vanishes; this is the null-quadrature form of the one-phase problem and, for smooth boundaries, is equivalent to

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = \partial_\nu u = 0 \quad \text{on } \partial\Omega, \qquad u = 0 \quad \text{on } \mathbb{C} \setminus \Omega.$$

This is the local Cauchy-transform version of the perturbative free-boundary method of Isakov [23, Chapter 5], and is closely related to the quadrature-domain constructions of [13, 22, 39, 38, 24].

Let $\Omega_0 \subset \mathbb{C} \simeq \mathbb{R}^2$ be a bounded simply connected C^2 domain. For $\rho > 0$ sufficiently small, every C^2 domain Ω which is ρ-close to Ω_0 can be written uniquely as a normal graph over $\partial\Omega_0$:

$$\partial\Omega = \{x + \varphi(x)\nu(x) : x \in \partial\Omega_0\},$$

where ν is the outer unit normal and $\varphi \in C^2(\partial\Omega_0)$ is small. We denote this domain by Ω_φ.

Proposition 2.1 (Perturbative null-quadrature theorem)**.** *Let $\Omega_0 \subset \mathbb{C}$ be a bounded simply connected C^2 domain, let Ω' be an open neighborhood of $\overline{\Omega_0}$, and let $f_0 \in C^5(\overline{\Omega'})$ satisfy*

$$f_0 > 0 \qquad \text{on } \partial\Omega_0.$$

Assume that there exists a compactly supported function

$$u_0 \in C^1(\mathbb{C}) \cap C^{2,\alpha}(\overline{\Omega_0}) \qquad \text{for some } \alpha \in (0,1)$$

such that

$$\begin{aligned} -\Delta u_0 &= f_0 \, \mathbf{1}_{\Omega_0} \qquad \text{in } \mathcal{D}'(\mathbb{C}), \\ u_0 = 0 \quad \text{on } \mathbb{C} \setminus \Omega_0, &\qquad u_0 < 0 \quad \text{in } \Omega_0. \end{aligned} \tag{2.8}$$

Then there exist $\varepsilon > 0$ and $\rho > 0$ such that, whenever

$$\|f - f_0\|_{C^5(\overline{\Omega'})} < \varepsilon,$$

there are a unique normal graph

$$\Omega_f = \Omega_{\varphi(f)}, \qquad \varphi(f) \in C^2(\partial\Omega_0), \qquad \|\varphi(f)\|_{C^2} < \rho,$$

and a compactly supported function

$$u_f \in C^1(\mathbb{C}) \cap C^{2,\alpha}(\overline{\Omega_f})$$

such that

$$\begin{aligned} -\Delta u_f &= f \, \mathbf{1}_{\Omega_f} \qquad \text{in } \mathcal{D}'(\mathbb{C}), \\ u_f = 0 \quad \text{on } \mathbb{C} \setminus \Omega_f, &\qquad u_f < 0 \quad \text{in } \Omega_f. \end{aligned} \tag{2.9}$$

In particular,

$$u_f = \partial_\nu u_f = 0 \qquad \text{on } \partial\Omega_f.$$

Moreover, $f \mapsto \varphi(f)$ is C^1 from a neighborhood of f_0 in $C^5(\overline{\Omega'})$ into $C^2(\partial\Omega_0)$.

Proof. We prove the result by imposing the vanishing of the exterior Cauchy transform as the free-boundary equation.

Step 1: the Cauchy-transform equation. Choose a bounded C^2 domain Ω_2 such that

$$\overline{\Omega_0} \subset \Omega_2, \qquad \overline{\Omega_2} \subset \Omega',$$

and restrict to normal graphs with $\overline{\Omega_\varphi} \subset \Omega_2$. For such φ, define

$$\mathcal{C}(\varphi, f)(z) := \int_{\Omega_\varphi} \frac{f(w)}{z-w}\, dA(w), \qquad z \in \partial\Omega_2.$$

If $\mathcal{C}(\varphi, f)$ vanishes on $\partial\Omega_2$, then it vanishes on $\mathbb{C} \setminus \overline{\Omega_\varphi}$ by holomorphic continuation, since it is holomorphic there and vanishes at infinity.

For the reference pair this condition holds. Indeed, from $-\Delta u_0 = f_0 \mathbf{1}_{\Omega_0}$ and $u_0 = 0$ on $\mathbb{C} \setminus \Omega_0$, the function $\partial_z u_0$ vanishes outside Ω_0 and solves

$$\partial_{\bar z}(\partial_z u_0) = -\frac{1}{4} f_0\, \mathbf{1}_{\Omega_0}.$$

The Cauchy transform of $f_0 \mathbf{1}_{\Omega_0}$ solves the same $\bar\partial$-equation up to the universal constant and vanishes at infinity. Their difference is entire and vanishes at infinity, hence is zero. Thus

$$\mathcal{C}(0, f_0) = 0 \qquad \text{on } \partial\Omega_2.$$

Step 2: smooth dependence and linearization. Let Y be the Banach space of traces on $\partial\Omega_2$ of exterior Cauchy transforms of C^2 densities on $\partial\Omega_0$:

$$Y := \left\{ z \mapsto \int_{\partial\Omega_0} \frac{\sigma(w)}{z-w}\, d\sigma(w)\Big|_{\partial\Omega_2} : \ \sigma \in C^2(\partial\Omega_0) \right\}.$$

We equip Y with the quotient norm inherited from $C^2(\partial\Omega_0)$; by the exterior Cauchy inversion theorem this is equivalent to the C^2-norm of the exterior holomorphic representative on any fixed annulus between $\partial\Omega_0$ and $\partial\Omega_2$. The standard exterior balayage property for the Cauchy transform shows that $\mathcal{C}(\varphi, f)|_{\partial\Omega_2} \in Y$: any C^2 volume density supported in Ω_2 has, in the exterior of Ω_2, the same Cauchy transform as a unique C^2 density on $\partial\Omega_0$.

The map

$$\mathcal{C} : C^2(\partial\Omega_0) \times C^5(\overline{\Omega'}) \to Y$$

is C^1 near $(0, f_0)$. This follows by writing $\Omega_\varphi = \Phi_\varphi(\Omega_0)$:

$$\mathcal{C}(\varphi, f)(z) = \int_{\Omega_0} \frac{f(\Phi_\varphi(w)) J_\varphi(w)}{z - \Phi_\varphi(w)}\, dA(w), \qquad z \in \partial\Omega_2,$$

where the denominator stays uniformly away from zero.

For $\psi \in C^2(\partial\Omega_0)$, the Hadamard variation formula gives

$$D_\varphi \mathcal{C}(0, f_0)[\psi](z) = \int_{\partial\Omega_0} \frac{f_0(w)\psi(w)}{z-w}\, d\sigma(w), \qquad z \in \partial\Omega_2. \tag{2.10}$$

Call this operator L.

Step 3: invertibility of the linearized operator. If $L\psi = 0$ on $\partial\Omega_2$, then the Cauchy transform of the boundary density $f_0\psi$ vanishes on the exterior of Ω_0 by the identity theorem. Taking boundary values on $\partial\Omega_0$ and using the Sokhotski–Plemelj jump relation gives $f_0\psi = 0$, hence $\psi = 0$ because $f_0 > 0$ on $\partial\Omega_0$. Thus L is injective.

Conversely, if $h \in Y$, let H be its holomorphic extension to $\mathbb{C} \setminus \overline{\Omega_0}$, vanishing at infinity. The exterior Cauchy inversion problem on the C^2 curve $\partial\Omega_0$ gives a unique density $\sigma \in C^2(\partial\Omega_0)$ such that

$$H(z) = \int_{\partial\Omega_0} \frac{\sigma(w)}{z-w}\, d\sigma(w), \qquad z \in \mathbb{C} \setminus \overline{\Omega_0}.$$

Setting $\psi = \sigma/f_0$ gives $L\psi = h$. The Fredholm theory of the Cauchy singular integral on C^2 curves gives continuous dependence of σ, hence L^{-1} is bounded.

Step 4: implicit function theorem and construction of u_f. Since $\mathcal{C}(0, f_0) = 0$ and $D_\varphi \mathcal{C}(0, f_0) = L$ is an isomorphism, the implicit function theorem gives a unique small $\varphi(f)$ solving

$$\mathcal{C}(\varphi(f), f) = 0 \qquad \text{on } \partial\Omega_2.$$

Let $\Omega_f = \Omega_{\varphi(f)}$. By holomorphic continuation,

$$\int_{\Omega_f} \frac{f(w)}{z-w}\, dA(w) = 0 \qquad \text{for } z \in \mathbb{C} \setminus \overline{\Omega_f}.$$

Define

$$N_f(z) := -\frac{1}{2\pi} \int_{\Omega_f} \log|z-w|\, f(w)\, dA(w).$$

Then $-\Delta N_f = f\mathbf{1}_{\Omega_f}$, and the vanishing of the Cauchy transform says that $\partial_z N_f = 0$ on the exterior. Thus N_f is constant on $\mathbb{C} \setminus \overline{\Omega_f}$. Subtracting this constant gives a compactly supported solution u_f of (2.9), with $u_f = 0$ on the exterior. Regularity of logarithmic potentials with bounded compactly supported density gives $u_f \in C^1(\mathbb{C})$, while interior Schauder estimates give $u_f \in C^{2,\alpha}(\overline{\Omega_f})$. Since $u_f \equiv 0$ outside, also $\partial_\nu u_f = 0$ on $\partial\Omega_f$.

Step 5: preservation of negativity. On compact subsets of Ω_0, the pullbacks $u_f \circ \Phi_{\varphi(f)}$ converge uniformly to u_0, so strict negativity persists away from the boundary. Near the boundary, use the second-order nondegeneracy. Since $u_f = \partial_\nu u_f = 0$ on $\partial\Omega_f$ and $-\Delta u_f = f$, while all tangential derivatives of the boundary trace vanish, one has

$$\partial_{\nu\nu} u_f = -f \qquad \text{on } \partial\Omega_f.$$

Because $f_0 > 0$ on $\partial\Omega_0$, after shrinking ε there is $c_0 > 0$ such that $\partial_{\nu\nu} u_f \le -c_0$ on $\partial\Omega_f$. Taylor expansion along inward normal segments gives

$$u_f(x - t\nu_f(x)) \le -\frac{c_0}{4} t^2 < 0$$

for $0 < t < t_0$, uniformly in $x \in \partial\Omega_f$. Thus $u_f < 0$ in a boundary collar and hence throughout Ω_f. The uniqueness and C^1 dependence of $\varphi(f)$ are those supplied by the implicit function theorem. □

2.3.2. *A perturbative weighted quadrature-domain theorem.* We now prove a local realization statement for weighted quadrature domains near a reference domain.

Fix a positive weight $\mu \in C^2(\overline{U_2})$, where $U_1 \subset U_0 \subset U_2$ are bounded C^2 domains. For a bounded domain $U \subset U_2$, define its weighted Cauchy transform by

$$g_U(z) := \int_U \frac{\mu(w)}{z-w}\, dA(w), \qquad z \in \mathbb{C} \setminus \overline{U}.$$

This defines a holomorphic function on $\mathbb{C} \setminus \overline{U}$, and, in the sense of distributions on $\mathbb{C}$,

$$\partial_{\bar{z}} g_U = \pi\, \mu\, \mathbf{1}_U.$$

As before, every domain U sufficiently close to U_0 may be written as a normal graph over ∂U_0:

$$\partial U = \{x + \varphi(x)\nu(x) : x \in \partial U_0\},$$

with $\varphi \in C^2(\partial U_0)$ small; we denote this domain by U_φ.

Proposition 2.2 ([13])**.** *Let $U_1 \subset U_0 \subset U_2$ be bounded C^2 domains, and let $\mu \in C^2(\overline{U_2})$ satisfy*

$$\mu > 0 \qquad \text{on } \overline{U_2}.$$

Then there exist $\rho > 0$ and $\varepsilon > 0$ with the following property.

Suppose g is holomorphic on $\mathbb{C} \setminus \overline{U_1}$, continuous up to ∂U_1, and satisfies

$$\lim_{z\to\infty} g(z) = 0, \qquad \lim_{z\to\infty} z g(z) = c_0 > 0, \tag{2.11}$$

together with

$$\sup_{z\in\partial U_1} |g(z) - g_{U_0}(z)| < \varepsilon.$$

Then there exists a unique C^2 function $\varphi \in C^2(\partial U_0)$, with $\|\varphi\|_{C^2} < \rho$, such that the associated domain U_φ satisfies

$$U_1 \subset U_\varphi \subset U_2$$

and

$$g_{U_\varphi}(z) = g(z) \qquad \text{for all } z \in \mathbb{C} \setminus \overline{U_\varphi}.$$

Moreover, the map

$$g \mapsto \varphi(g)$$

is C^1 with respect to the $C^2(\partial U_1)$ norm on the trace data and the $C^2(\partial U_0)$ norm on the boundary graph.

Proof. The proof is again an application of the implicit function theorem.

Step 1: reduction to a boundary matching problem. Since both g and g_{U_φ} are holomorphic on $\mathbb{C} \setminus \overline{U_1}$ and vanish at infinity, it is enough to impose equality on the boundary ∂U_1. More precisely, if

$$g_{U_\varphi} = g \qquad \text{on } \partial U_1,$$

then by the identity theorem the equality extends to all of $\mathbb{C} \setminus \overline{U_\varphi}$, because the difference is holomorphic on the common exterior domain and vanishes on the closed curve ∂U_1, which lies strictly inside that domain once $U_1 \subset U_\varphi$. Equivalently, since the zero set of a nontrivial holomorphic function is discrete, vanishing on the one-dimensional set ∂U_1 forces the difference to be identically zero on the connected component of $\mathbb{C} \setminus \overline{U_\varphi}$ containing infinity.

Thus we define

$$\mathcal{G}(\varphi)(z) := g_{U_\varphi}(z)\big|_{\partial U_1}, \qquad z \in \partial U_1.$$

We regard $\mathcal{G}$ as a map from a neighborhood of 0 in $C^2(\partial U_0)$ into $C^2(\partial U_1)$ (or, equivalently, into the space of traces on ∂U_1 of holomorphic functions on $\mathbb{C} \setminus \overline{U_1}$ vanishing at infinity).

The dependence of g_{U_φ} on φ is C^1: indeed, writing U_φ as the image of U_0 under the graph diffeomorphism Φ_φ, one obtains

$$g_{U_\varphi}(z) = \int_{U_0} \frac{\mu(\Phi_\varphi(w))J_\varphi(w)}{z - \Phi_\varphi(w)}\, dA(w),$$

and differentiation with respect to φ is justified by the smoothness of μ, the smallness of the perturbation, and the fact that $z \in \partial U_1$ stays a positive distance from $\overline{U_0}$.

Step 2: computation of the linearization. Let $\psi \in C^2(\partial U_0)$. The first variation of the weighted Cauchy transform with respect to a normal deformation of the domain is

$$D\mathcal{G}(0)[\psi](z) = \int_{\partial U_0} \frac{\mu(w)\psi(w)}{z - w}\, d\sigma(w), \qquad z \in \partial U_1. \tag{2.12}$$

This is the standard Hadamard variation formula: to first order, varying the domain in the normal direction by ψ replaces the bulk integral by a boundary layer integral with density $\mu\psi$.

Therefore the linearized operator

$$L\psi(z) := \int_{\partial U_0} \frac{\mu(w)\psi(w)}{z - w}\, d\sigma(w), \qquad z \in \partial U_1,$$

is precisely the weighted Cauchy transform of the boundary density $\mu\psi$.

Step 3: invertibility of the linearized operator. We claim that L is injective. Suppose $L\psi = 0$ on ∂U_1. Since $L\psi$ is holomorphic on $\mathbb{C} \setminus \overline{U_0}$, vanishes at infinity, and vanishes on the closed curve ∂U_1 contained in that exterior region, the identity theorem yields

$$L\psi \equiv 0 \qquad \text{on } \mathbb{C} \setminus \overline{U_0}.$$

Taking non-tangential boundary values on ∂U_0 and using the standard jump relation for the Cauchy transform, we conclude that

$$\mu\psi = 0 \qquad \text{on } \partial U_0.$$

Since $\mu > 0$ on ∂U_0, it follows that $\psi = 0$. Thus L is injective.

To see surjectivity onto the natural trace space, let h be the trace on ∂U_1 of a holomorphic function on $\mathbb{C} \setminus \overline{U_1}$ which vanishes at infinity and has the admissible principal part there. Since $\partial U_0 \subset \mathbb{C} \setminus \overline{U_1}$, the same function is holomorphic in a full neighborhood of ∂U_0. Denote by H this holomorphic function on $\mathbb{C} \setminus \overline{U_1}$. By the Cauchy integral formula applied on ∂U_1, the values of H on the annular region $\mathbb{C} \setminus (\overline{U_0} \cup U_1)$ are uniquely determined by its trace h. In particular, the trace $H|_{\partial U_0}$ depends continuously on h.

Now apply the standard Sokhotski–Plemelj inversion on ∂U_0: there exists a unique density $\sigma \in C^2(\partial U_0)$ such that

$$H(z) = \int_{\partial U_0} \frac{\sigma(w)}{z - w}\, d\sigma(w), \qquad z \in \mathbb{C} \setminus \overline{U_0},$$

and σ depends continuously on the boundary trace $H|_{\partial U_0}$. Since $\mu > 0$ on ∂U_0, setting

$$\psi := \frac{\sigma}{\mu}$$

gives $\psi \in C^2(\partial U_0)$ and

$$L\psi(z) = H(z) \qquad \text{for } z \in \partial U_1.$$

Thus L is onto the trace space under consideration. Combining injectivity and surjectivity, L is an isomorphism between $C^2(\partial U_0)$ and that trace space. In particular, after fixing any convenient Banach norm on the target, L^{-1} is bounded by the open mapping theorem.

Step 4: application of the implicit function theorem. Since $\mathcal{G}$ is C^1 and $D\mathcal{G}(0) = L$ is invertible, the implicit function theorem gives $\rho > 0$ and $\varepsilon > 0$ such that, for every g with

$$\|g - g_{U_0}\|_{C^2(\partial U_1)} < \varepsilon,$$

there exists a unique $\varphi \in C^2(\partial U_0)$, $\|\varphi\|_{C^2} < \rho$, such that

$$\mathcal{G}(\varphi) = g|_{\partial U_1}.$$

Moreover, the implicit function theorem yields that the solution map

$$g \mapsto \varphi(g)$$

is C^1 on a neighborhood of $g_{U_0}|_{\partial U_1}$ in the trace space. By construction,

$$g_{U_\varphi} = g \qquad \text{on } \partial U_1,$$

hence on the whole common exterior domain by holomorphic continuation.

Finally, by taking ρ smaller if necessary, every such U_φ still satisfies

$$U_1 \subset U_\varphi \subset U_2.$$

This proves the proposition. □

Remark 2.1. The only point in Proposition 2.2 that is slightly nontrivial is the invertibility of the linearized map. In the present perturbative regime, the key observation is that the Cauchy transform of a boundary density is determined by its trace on any inner contour, and the standard jump relation then recovers the density itself. This is exactly the mechanism needed in the proof of Theorem 1.

We postpone the proof of Theorem 2 until after the perturbative inverse theorem. The same inverse/free-boundary mechanism used to construct domains for prescribed eigenfunctions also recovers the classical Abreu–Dörfler disk theorem in the simply connected setting; see Section 4 below.

3. Proof of Theorem 1

Proof of Theorem 1. We again use the Bargmann transform. Let

$$P(z) = 1 + \sum_{k=1}^{N} b_k z^k,$$

where the coefficients b_k are chosen so that the Bargmann transform of

$$f_0 = h_0 + \sum_{k=1}^{N} a_k h_k$$

is exactly P. Then the condition that f_0 be an eigenfunction of $\mathcal{L}_U$ with eigenvalue λ is equivalent to

$$\int_U P(z) e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda P(w), \qquad w \in \mathbb{C}. \tag{3.1}$$

Step 0: a first reduction. For a polynomial

$$R(z) = \alpha_0 + \alpha_1 z + \cdots + \alpha_n z^n,$$

define the differential operator

$$R(\partial_w/\pi) := \sum_{k=0}^{n} \alpha_k \Big(\frac{\partial_w}{\pi}\Big)^k.$$

Applying $P(\partial_w/\pi)$ to both sides of (3.1), and using

$$\Big(\frac{\partial_w}{\pi}\Big)^j e^{\pi \bar{z} w} = \bar{z}^j e^{\pi \bar{z} w},$$

which is legitimate because $P(\partial_w/\pi)$ is a finite-order differential operator and the left-hand side of (3.1) is an entire function of w, we obtain

$$\int_U |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda Q(w), \qquad w \in \mathbb{C}, \tag{3.2}$$

where

$$Q(w) := P(\partial_w/\pi)\big(P(w)\big).$$

Thus, our first task is to construct a domain U satisfying (3.2).

Step 1: construction of an auxiliary domain. We first show that there exists a C^2 domain U_0 such that

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}. \tag{3.3}$$

Let $\psi \in C_c^\infty(\mathbb{C})$ be a nonnegative radial function such that

$$\operatorname{supp} \psi \subset D_2(0) \setminus D_1(0),$$

and define

$$\psi_\delta(z) := \delta^{-2} \psi(z/\delta).$$

Since ψ_δ is radial, a polar-coordinate computation shows that

$$\int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = c_\delta,$$

where

$$c_\delta := \int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2}\, dA(z).$$

Thus (3.3) is equivalent to

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z) = \frac{\lambda}{c_\delta} \int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z). \tag{3.4}$$

We shall construct directly a compactly supported distributional solution of the Poisson equation

$$\Delta u_0 = -|P(z)|^2 e^{-\pi|z|^2} \mathbf{1}_{U_0} + \frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}. \tag{3.5}$$

Once such a solution has been constructed, the moment identity (3.3) follows by applying the same Fourier-analytic divisibility argument used in the proof of Theorem 2 to the corresponding compactly supported distribution

$$\omega_0 := |P(z)|^2 e^{-\pi|z|^2} \mathbf{1}_{U_0} - \frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}.$$

Indeed, (3.5) says exactly that $\Delta u_0 = -\omega_0$, so the vanishing of the Fourier transform of ω_0 on the two characteristic lines is equivalent to (3.3).

We first consider the unperturbed case $P \equiv 1$. By the theorem of Daubechies [15], there exists a disk D_0 such that

$$\int_{D_0} e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}.$$

Hence there exists a compactly supported solution $\widetilde{u}_0$ of

$$\Delta \widetilde{u}_0 = -e^{-\pi|z|^2} \mathbf{1}_{D_0} + \frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}. \tag{3.6}$$

If $\delta > 0$ is sufficiently small, then $\operatorname{supp} \psi_\delta \subset D_0$. Since all data are radial, the solution $\widetilde{u}_0$ is radial as well. We claim that, for δ sufficiently small,

$$\widetilde{u}_0 > 0 \qquad \text{in } D_0.$$

Indeed, let u_* denote the compactly supported radial solution of

$$\Delta u_* = -e^{-\pi|z|^2} \mathbf{1}_{D_0} + \lambda \delta_0.$$

By the defining property of D_0, this is exactly the reference obstacle-type potential associated with the disk, and $u_* > 0$ in D_0: away from the origin, $-\Delta u_* = e^{-\pi|z|^2} > 0$ in $D_0 \setminus \{0\}$, $u_* = 0$ on ∂D_0, and the logarithmic singularity created by the mass $\lambda\delta_0$ is positive. On the other hand, the family

$$\frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}$$

converges to $\lambda\delta_0$ in the sense of distributions as $\delta \to 0$, so the corresponding compactly supported radial solutions $\widetilde{u}_0$ converge locally uniformly on $\overline{D}_0 \setminus \{0\}$ to u_*. Since u_* is strictly positive on compact subsets of $D_0 \setminus \{0\}$, and the support of ψ_δ lies in an arbitrarily small neighborhood of the origin, it follows that $\widetilde{u}_0 > 0$ throughout D_0 for $\delta > 0$ sufficiently small.

We shall now invoke the perturbative free-boundary theorem of Isakov, given by Proposition 2.1, verifying its hypotheses carefully. Set

$$f_0(z) := e^{-\pi|z|^2} \Big(1 - \frac{\lambda}{c_\delta} \psi_\delta(z)\Big).$$

Since $\psi_\delta \in C_c^\infty(\mathbb{C})$, we have $f_0 \in C^\infty(\mathbb{R}^2)$, hence in particular $f_0 \in C^5(\mathbb{R}^2)$. If δ is chosen so small that $\operatorname{supp}\psi_\delta \subset D_0$, then on ∂D_0 one has

$$f_0(z) = e^{-\pi|z|^2} > 0.$$

Next, define

$$v_0 := -\widetilde{u}_0.$$

Then (3.6) becomes

$$-\Delta v_0 = f_0\,\mathbf{1}_{D_0} \qquad \text{in } \mathbb{C},$$

with $v_0 = 0$ on $\mathbb{C} \setminus D_0$ and $v_0 < 0$ in D_0. Since $\widetilde{u}_0$ was constructed as the compactly supported radial solution of (3.6), it is C^1 across ∂D_0; equivalently $v_0 = \partial_\nu v_0 = 0$ on ∂D_0. Thus the hypotheses of Proposition 2.1 are satisfied with $\Omega_0 = D_0$. Now define

$$f(z) := e^{-\pi|z|^2}\Big(|P(z)|^2 - \frac{\lambda}{c_\delta}\psi_\delta(z)\Big).$$

Since P is a polynomial, $f \in C^\infty(\mathbb{R}^2)$ as well. Moreover, for any fixed disk $\Omega' \supset D_0$, the $C^5(\Omega')$ norm of $f - f_0$ is controlled by the coefficients of $P - 1$:

$$\|f - f_0\|_{C^5(\Omega')} \le C(\Omega', N, \lambda) \max_{1\le k\le N} |b_k|.$$

Hence, if the coefficients b_k are sufficiently small, we may arrange that

$$\|f - f_0\|_{C^5(\Omega')} < \varepsilon_1.$$

Applying Proposition 2.1, we obtain a C^2 domain U_0, represented as a small normal deformation of D_0, and a compactly supported function v such that

$$-\Delta v = f\,\mathbf{1}_{U_0} \qquad \text{in } \mathbb{C},$$

with $v = 0$ on $\mathbb{C} \setminus U_0$, $\partial_\nu v = 0$ on ∂U_0, and $v < 0$ in U_0. Set

$$u_0 := -v.$$

Then

$$\Delta u_0 = f\,\mathbf{1}_{U_0}.$$

If the perturbation is sufficiently small, then U_0 remains a small deformation of D_0, and since $\operatorname{supp}\psi_\delta \subset D_0$ we still have

$$\operatorname{supp}\psi_\delta \subset U_0.$$

Therefore

$$\Delta u_0 = -|P(z)|^2 e^{-\pi|z|^2}\mathbf{1}_{U_0} + \frac{\lambda}{c_\delta}\psi_\delta(z)e^{-\pi|z|^2},$$

so (3.5) holds. Hence (3.3) is proved.

Step 2: construction of a domain satisfying (3.2). We now convert (3.2) into the weighted Cauchy-transform form required by Proposition 2.2. Restrict (3.2) to real values $w = t > 0$, multiply by $e^{-\pi\zeta t}$, and integrate in $t > 0$. If

$$\zeta > \sup_{z\in U} |z|,$$

Fubini's theorem gives

$$\int_U \left(\int_0^\infty e^{\pi(\overline{z}-\zeta)t}\, dt \right) |P(z)|^2 e^{-\pi|z|^2}\, dA(z) = \lambda \int_0^\infty e^{-\pi\zeta t} Q(t)\, dt.$$

Indeed, for real $\zeta > \sup_{z\in U} |z|$ one has

$$\Re(\overline{z}-\zeta) \le |z| - \zeta < 0 \qquad \text{for every } z \in U,$$

so the inner exponential is integrable uniformly in $z \in U$. Writing

$$Q(w) = \alpha_0 + \alpha_1 w + \cdots + \alpha_n w^n,$$

we obtain

$$\frac{1}{\pi} \int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta - \overline{z}}\, dA(z) = \lambda \sum_{k=0}^n \alpha_k \int_0^\infty e^{-\pi\zeta t} t^k\, dt.$$

Since

$$\int_0^\infty e^{-\pi\zeta t} t^k\, dt = \frac{\Gamma(k+1)}{(\pi\zeta)^{k+1}},$$

it follows that

$$\frac{1}{\pi} \int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta - \overline{z}}\, dA(z) = \lambda \sum_{k=0}^n \alpha_k \frac{\Gamma(k+1)}{\pi^{k+1}\zeta^{k+1}}.$$

To put this into the form of a weighted Cauchy transform, let

$$V := \overline{U}, \qquad \mu(w) := |P(\overline{w})|^2 e^{-\pi|w|^2}.$$

Changing variables $w = \overline{z}$ yields

$$\int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta - \overline{z}}\, dA(z) = \int_V \frac{\mu(w)}{\zeta - w}\, dA(w) = g_V(\zeta).$$

Hence

$$g_V(\zeta) = \lambda \sum_{k=0}^n \alpha_k \frac{\Gamma(k+1)}{\pi^k \zeta^{k+1}}, \qquad \zeta \in \mathbb{R}_+,\ \zeta \gg 1. \tag{3.7}$$

Since both sides are holomorphic for large ζ, analytic continuation gives the desired identity on $\mathbb{C} \setminus V$.

Conversely, if a domain V satisfies (3.7), then by expanding at infinity and comparing coefficients one recovers the moment identities corresponding to (3.2) for the conjugate domain $U = \overline{V}$. Thus the conjugation is merely a bookkeeping device that places the kernel in the form $1/(z-w)$ required by the weighted Cauchy transform formalism.

Let us denote by g the rational function on the right-hand side of (3.7). We now verify the hypotheses of Proposition 2.2. Since

$$Q(w) = P(\partial_w/\pi)\big(P(w)\big),$$

and $P(0) = 1$, the constant term of Q is

$$Q(0) = 1 + \sum_{k=1}^N \frac{k!}{\pi^k} b_k^2.$$

In particular, if the coefficients b_k are sufficiently small, then

$$Q(0) > 0.$$

Therefore the leading term in the expansion of g at infinity is

$$\frac{\lambda Q(0)}{z},$$

so

$$g(z) \to 0, \qquad z\, g(z) \to \lambda Q(0) > 0 \qquad \text{as } z \to \infty.$$

Thus (2.11) holds, with $c_0 = \lambda Q(0)$.

We next verify the closeness condition on ∂U_1. Since U_0 was obtained in Step 1 as a small perturbation of the Daubechies disk, we choose $U_1 \subset U_0 \subset U_2$ to be concentric disks with radii, say, $(1 - 10^{-3})$ and $(1 + 10^{-3})$ times the radius of D_0. By Step 1,

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}.$$

Repeating the Laplace-transform argument above therefore gives

$$g_{U_0}(z) = \frac{\lambda}{z} \qquad \text{for } z \in \mathbb{C} \setminus \overline{U_0},$$

where g_{U_0} denotes the weighted Cauchy transform of the conjugate domain $\overline{U_0}$ with weight $\mu(w) = |P(\overline{w})|^2 e^{-\pi|w|^2}$. Consequently, on ∂U_1,

$$g(z) - g_{U_0}(z) = \lambda \sum_{k=1}^{n} \alpha_k \frac{\Gamma(k+1)}{\pi^k z^{k+1}} + \lambda\,(Q(0) - 1)\,\frac{1}{z}.$$

Since ∂U_1 is a fixed circle staying a positive distance from the origin, it follows that

$$\sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)| \le C(U_1, \lambda)\Big(|Q(0) - 1| + \sum_{k=1}^{n} |\alpha_k|\Big).$$

Each coefficient α_k of $Q - 1$ depends polynomially on the coefficients of $P - 1$, and therefore tends to zero with $\max_{1 \le j \le N} |b_j|$. Hence, by choosing the coefficients sufficiently small, we may ensure that

$$\sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)| < \varepsilon_0.$$

Therefore Proposition 2.2 applies and yields a C^2 domain $\widetilde{U}$ such that

$$g_{\widetilde{U}}(z) = g(z), \qquad z \in \mathbb{C} \setminus \widetilde{U}.$$

Replacing $\widetilde{U}$ by its conjugate if necessary, we obtain a domain U satisfying (3.2).

Finally, since $U \subset U_2$ and U_2 is fixed once λ is fixed, by shrinking the admissible size of the coefficients of $P - 1$ once more we may ensure that

$$P(z) \neq 0 \qquad \text{for all } z \in U_2.$$

In particular,

$$P(z) \neq 0 \qquad \text{for all } z \in U.$$

Step 3: recovery of the original eigenvalue equation. Let

$$G(w) := \int_U P(z)e^{-\pi|z|^2}e^{\pi\bar{z}w}\,dA(z).$$

By construction, (3.2) holds, so applying $P(\partial_w/\pi)$ to G gives

$$P(\partial_w/\pi)G(w) = P(\partial_w/\pi)(\lambda P)(w), \qquad w \in \mathbb{C}.$$

Hence

$$H := G - \lambda P$$

satisfies

$$P(\partial_w/\pi)H = 0.$$

Introduce the characteristic polynomial

$$\widetilde{P}(\zeta) := P(\zeta/\pi).$$

If $\zeta_1, \dots, \zeta_\ell$ are the zeros of $\widetilde{P}$, with multiplicities $m_1, \dots, m_\ell$, then standard ODE theory implies that every entire solution of

$$P(\partial_w/\pi)H = 0$$

of at most exponential growth is of the form

$$H(w) = \sum_{j=1}^{\ell} R_j(w)e^{\zeta_j w},$$

where each R_j is a polynomial of degree strictly less than m_j. Since P has no zeros on U, and in fact none on the larger fixed set U_2, by choosing the coefficients of $P-1$ sufficiently small we may also assume that all zeros of $\widetilde{P}$ stay at distance at least $10\sup_{z\in U}|z|$ from the origin:

$$|\zeta_j| > 10\sup_{z\in U}|z|, \qquad 1 \le j \le \ell.$$

Suppose, for contradiction, that some R_j is not identically zero. Reorder so that

$$|\zeta_1| = \max_j |\zeta_j| \qquad \text{and} \qquad R_1 \not\equiv 0.$$

Then along the ray

$$w = t\,\overline{\zeta_1}, \qquad t > 0,$$

the dominant exponential term yields

$$|H(t\overline{\zeta_1})| \ge c\,e^{t|\zeta_1|^2}$$

for all sufficiently large t. On the other hand, since P is a fixed polynomial and U is bounded, if

$$M := \sup_{z\in U}|z|,$$

there exists $C > 0$ such that

$$|G(w)| \le C(1+|w|)^N e^{\pi M|w|}, \qquad w \in \mathbb{C}.$$

Indeed,

$$|G(w)| \le \int_U |P(z)| e^{-\pi|z|^2} e^{\pi|z||w|}\, dA(z) \le C(1+|w|)^N e^{\pi M|w|},$$

because P has degree at most N and U is bounded. Since every polynomial is dominated by $e^{\varepsilon|w|}$ for arbitrary $\varepsilon > 0$, it follows that for every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$|G(w)| \le C_\varepsilon e^{(\pi+\varepsilon)M|w|}, \qquad w \in \mathbb{C}.$$

As $\lambda P(w)$ grows only polynomially, the same estimate holds for H:

$$|H(w)| \le C'_\varepsilon e^{(\pi+\varepsilon)M|w|}.$$

Evaluating at $w = t\overline{\zeta_1}$ gives

$$|H(t\overline{\zeta_1})| \le C'_\varepsilon e^{(\pi+\varepsilon)Mt|\zeta_1|}.$$

Comparing the two inequalities and letting $t \to \infty$, we obtain

$$|\zeta_1| \le (\pi+\varepsilon)M.$$

Since $\varepsilon > 0$ is arbitrary, this implies

$$|\zeta_1| \le \pi M,$$

contradicting our choice

$$|\zeta_1| > 10M.$$

Hence every R_j vanishes identically, so

$$H \equiv 0.$$

Equivalently,

$$G = \lambda P,$$

which is exactly (3.1). This proves that P, and therefore f_0, is an eigenfunction of the localization operator associated with the domain U with eigenvalue λ. □

4. A Classical Rigidity Corollary

4.1. **First proof.**

First proof under the additional assumption that ∂U is C^2. The Bargmann transform of a Hermite function h_k is a nonzero multiple of the monomial z^k. Thus, arguing as in [1], the assumption that h_k is an eigenfunction of the localisation operator $\mathcal{L}_U$ is equivalent to the identity

$$\int_U z^k e^{\pi w\overline{z}} e^{-\pi|z|^2}\, dA(z) = \lambda w^k, \qquad w \in \mathbb{C}.$$

Both sides are entire functions of w. Differentiating k times with respect to w, we obtain

$$\int_{\mathbb{C}} \mathbf{1}_U(z)\, |z|^{2k} e^{-\pi|z|^2} e^{\pi\overline{z}w}\, dA(z) = c(\lambda, k), \tag{4.1}$$

for some real constant $c(\lambda, k)$.

Consider now the compactly supported distribution

$$\eta := |z|^{2k} e^{-\pi|z|^2} \mathbf{1}_U(z) - c(\lambda, k)\delta_0.$$

Writing $z = x + iy$, we may rewrite the exponential factor as

$$e^{\pi \bar{z} w} = e^{\pi (x-iy)w} = e^{\pi x w} e^{-i\pi y w}.$$

Hence (4.1) implies that the Fourier transform $\widehat{\eta}$, viewed as an entire function on $\mathbb{C}^2$ by Paley–Wiener, satisfies

$$\widehat{\eta}(w, -iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Since $c(\lambda, k) \in \mathbb{R}$, taking complex conjugates in (4.1) after replacing w by $\overline{w}$ also yields

$$\widehat{\eta}(w, iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Applying Lemma 2.6 to the distribution η, we obtain a function $u \in L^2(\mathbb{R}^2)$ such that

$$\Delta u = f(|z|)\mathbf{1}_U - c(\lambda, k)\delta_0 \tag{4.2}$$

in the sense of distributions, where

$$f(|z|) = |z|^{2k} e^{-\pi |z|^2}.$$

Writing u explicitly as a logarithmic potential, we conclude that U satisfies

$$\int_U \log|z - z'|\, f(|z'|)\, dA(z') = c(\lambda, k) \log|z|, \qquad z \in \mathbb{C} \setminus (U \cup \{0\}). \tag{4.3}$$

We now reduce to the case $0 \notin U$. Indeed, if $0 \in U$, then for every sufficiently small $\rho > 0$ with $D_\rho(0) \subset U$, the potential generated by

$$f(|z|)\mathbf{1}_{D_\rho(0)}$$

is radial outside $D_\rho(0)$, hence equals a constant plus a multiple of $\log|z|$ there. Therefore, after subtracting such a disk from U, identity (4.3) is preserved outside the new set, up to replacing the constant $c(\lambda, k)$ by another real constant. Thus, without loss of generality, we may assume that

$$0 \notin U.$$

Moreover, $0 \notin \partial U$, because the left-hand side of (4.3) stays finite as $z \to 0$, whereas the right-hand side diverges.

Define

$$\Psi_U(z) := \int_U \log|z - z'|\, f(|z'|)\, dA(z').$$

We claim that Ψ_U is *locally radial*: for each $z_0 \in U$, there exists a ball $B(z_0) \subset U$ such that Ψ_U agrees on $B(z_0)$ with a radial function.

To prove this, let $B \supset U$ be a ball centered at 0, and define

$$\Phi(z) := \int_B \log|z - z'|\, f(|z'|)\, dA(z').$$

Since both B and the weight $f(|z'|)$ are radial, Φ is radial. In particular, $\nabla\Phi(z)$ is parallel to z for all $z \in \mathbb{R}^2$. Also, for $z \in B$,

$$\Delta\Phi(z) = c\, f(|z|)$$

for a suitable constant $c > 0$.

Now Ψ_U satisfies (4.3), so on ∂U its gradient is parallel to z by continuity. Therefore the harmonic function

$$v := \Psi_U - \Phi$$

satisfies

$$\nabla v(x) \text{ is parallel to } x, \qquad x \in \partial U.$$

The next lemma shows that this forces v, and hence Ψ_U, to be locally radial.

Lemma 4.1. *Let $V \subset \mathbb{R}^2$ be a domain, and suppose there exists a harmonic function $v \in C^1(\overline{V})$ such that $\nabla v(x)$ is parallel to x for every $x \in \partial V$. Then for each circle $S \subset \mathbb{R}^2$, the function v is constant on every connected component of $S \cap V$.*

Proof. Consider the angular derivatives

$$g_{i,j}(x) = x_i\, \partial_j v(x) - x_j\, \partial_i v(x).$$

Since v is harmonic, each $g_{i,j}$ is harmonic in V. By the boundary assumption,

$$g_{i,j} = 0 \qquad \text{on } \partial V.$$

Hence, by the maximum principle,

$$g_{i,j} \equiv 0 \qquad \text{in } V.$$

Therefore $\nabla v(x)$ is parallel to x for every $x \in V$. In particular, on any circle S, the tangential derivative of v vanishes on $S \cap V$, so v is constant on each connected component of $S \cap V$. □

Lemma 4.2. *Let $W \subset \mathbb{R}^2 \setminus \{0\}$ be a connected open set, and let $u \in C(W)$. Suppose that for every $x \in W$ there exist $r_x > 0$ and a scalar function ϕ_x such that*

$$u(y) = \phi_x(|y|) \qquad \textit{for all } y \in B(x, r_x) \subset W.$$

Then there exists a single scalar function ϕ, defined on the interval

$$I_W := \{|y| : y \in W\},$$

such that

$$u(y) = \phi(|y|) \qquad \textit{for all } y \in W.$$

Proof. Choose for each $x \in W$ a ball $B_x := B(x, r_x) \subset W$ on which u is a radial function of $|y|$. If $B_x \cap B_{x'} \neq \emptyset$, then

$$u(y) = \phi_x(|y|) = \phi_{x'}(|y|) \qquad \text{for every } y \in B_x \cap B_{x'}.$$

Since $B_x \cap B_{x'}$ is open and avoids the origin, the set of radii attained on $B_x \cap B_{x'}$ contains a nontrivial open interval. Hence ϕ_x and $\phi_{x'}$ agree on that interval.

Fix $x_* \in W$. For any $y \in W$, choose a polygonal path in W joining x_* to y. By compactness of the image of the path, it may be covered by finitely many balls

$$B_{x_1}, \ldots, B_{x_m}$$

with successive overlaps nonempty, $x_1 = x_*$, and $y \in B_{x_m}$. The previous paragraph shows that the corresponding local radial profiles agree on the overlaps, and therefore patch together

uniquely along the chain. This defines a single scalar function ϕ on I_W such that $u(z) = \phi(|z|)$ for every $z \in W$. The definition is independent of the chosen chain because any two chains can be concatenated and compared in the same way. $\square$

We now conclude the proof. Since U is simply connected, it suffices to rule out the possibility that U is nonradial. Suppose therefore that U is not a disk. Then U is neither a disk nor an annulus, so there exist radii $0 < r_1 < r_2$ such that the annulus

$$A := \{z \in \mathbb{R}^2 : \ r_1 < |z| < r_2\}$$

meets both U and ∂U. Choose a point

$$x_0 \in \partial U \cap A$$

and let

$$\widetilde{U}_0 := U \cap B_r(x_0),$$

for $r > 0$ sufficiently small. Shrinking r if necessary, we may assume that $\widetilde{U}_0$ is connected and that $0 \notin \overline{\widetilde{U}_0}$. Since Ψ_U is locally radial in U, Lemma 4.2 applied to $W = \widetilde{U}_0$ yields a scalar function ψ_U such that

$$\Psi_U(x) = \psi_U(|x|), \qquad x \in \widetilde{U}_0.$$

On the other hand, by (4.3), for every $z \in \partial\widetilde{U}_0$ we have

$$\nabla\Psi_U(z) = c\,\frac{z}{|z|^2}$$

for a suitable constant $c \in \mathbb{R}$. Since $\Psi_U(x) = \psi_U(|x|)$ on $\widetilde{U}_0$, this yields

$$\psi_U'(|z|)\frac{z}{|z|} = c\,\frac{z}{|z|^2}, \qquad z \in \partial\widetilde{U}_0,$$

hence

$$\psi_U'(r) = \frac{c}{r}$$

for all radii attained by points of $\partial\widetilde{U}_0$. Since ∂U is C^2 and $x_0 \in \partial U$, after shrinking r once more we may assume that the set of radii attained by $\partial\widetilde{U}_0 = \partial U \cap B_r(x_0)$ contains a nontrivial open interval $I \subset (0, \infty)$. Therefore

$$\psi_U'(r) = \frac{c}{r}$$

for every $r \in I$, whence

$$\psi_U(r) = c\log r + C_0 \qquad \text{for every } r \in I$$

for some constant C_0. Let

$$W_I := \{x \in \widetilde{U}_0 : \ |x| \in I\}.$$

Then W_I is a nonempty open subset of U, and on W_I we have

$$\Delta\Psi_U \equiv 0 \qquad \text{in } W_I.$$

This is impossible, because by construction

$$\Delta\Psi_U = c' f(|z|)$$

inside U, where $f(|z|) = |z|^{2k}e^{-\pi|z|^2}$ vanishes only at the origin, while $0 \notin \widetilde{U}_0$ and hence $0 \notin W_I$. Thus $\Delta\Psi_U$ cannot vanish on W_I, a contradiction. This contradiction shows that U must be radial. Since U is assumed bounded and simply connected, it follows that U is a disk. □

4.2. **Second proof.**

Second proof of Theorem 2. We keep the notation

$$f(|z|) := |z|^{2k}e^{-\pi|z|^2}, \qquad c := c(\lambda, k),$$

from the first proof. Starting again from (4.1) and applying Lemma 2.6, we obtain a compactly supported distributional solution of

$$\Delta u = f(|z|)\mathbf{1}_U - c\,\delta_0 \qquad \text{in } \mathcal{D}'(\mathbb{R}^2). \tag{4.4}$$

Using the planar fundamental solution $(2\pi)^{-1}\log|z|$, we may write

$$u(z) := \frac{1}{2\pi}\int_U \log|z - z'|\, f(|z'|)\, dA(z') - \frac{c}{2\pi}\log|z|. \tag{4.5}$$

By (4.3), this function satisfies

$$u \equiv 0 \qquad \text{on } \mathbb{R}^2 \setminus (U \cup \{0\}).$$

We first show that $0 \in U$. Since the singular term in (4.4) is $-c\delta_0$, one must have $0 \in \overline{U}$. If $0 \in \partial U$, then there exist exterior points $z_j \to 0$ with

$$u(z_j) = 0.$$

On the other hand, the first term in (4.5) remains finite as $z \to 0$, because $f \in L^\infty(U)$ and $\log|z - z'|$ is locally integrable, whereas the second term equals $-\frac{c}{2\pi}\log|z| \to +\infty$. This contradicts the exterior identity $u(z_j) = 0$. Hence $0 \notin \partial U$, and therefore

$$0 \in U.$$

We next show directly from (4.5) that

$$u \in C^1(\mathbb{R}^2 \setminus \{0\}).$$

Indeed, if we write

$$g(z) := f(|z|)\mathbf{1}_U(z) \in L^\infty_c(\mathbb{R}^2),$$

then

$$u(z) = \frac{1}{2\pi}\int_{\mathbb{R}^2} \log|z - z'|\, g(z')\, dA(z') - \frac{c}{2\pi}\log|z|.$$

Since the kernel $K(\zeta) = \zeta/|\zeta|^2$ belongs to $L^1_{\mathrm{loc}}(\mathbb{R}^2)$, the translation map $z \mapsto K(z - \cdot)$ is continuous in L^1_{loc}. It follows that

$$\nabla\left(\int_{\mathbb{R}^2} \log|z - z'|\, g(z')\, dA(z')\right) = \int_{\mathbb{R}^2} \frac{z - z'}{|z - z'|^2}\, g(z')\, dA(z')$$

defines a continuous function of $z \in \mathbb{R}^2$. Hence $u \in C^1(\mathbb{R}^2 \setminus \{0\})$.

Now let $z_* \in \partial U$. Since $0 \in U$, we have $z_* \neq 0$, and since

$$u \equiv 0 \qquad \text{on } \mathbb{R}^2 \setminus \overline{U},$$

both u and ∇u vanish on the exterior side of ∂U. By continuity of ∇u at z_*, we conclude that

$$\partial_z u = 0 \qquad \text{on } \partial U. \tag{4.6}$$

Define

$$\Phi(t) := \int_0^t s^k e^{-\pi s}\, ds \qquad \text{and} \qquad \Psi(z) := \frac{\Phi(|z|^2)}{z}, \quad z \neq 0.$$

Then

$$\partial_{\bar z}\Psi(z) = f(|z|) \qquad (z \neq 0),$$

and $\Phi(0) = 0$, so the same identity holds distributionally on U. Now set

$$F(z) := 4\partial_z u(z) - \Psi(z) + \frac{c}{\pi z}, \qquad z \in U \setminus \{0\}.$$

Using $\partial_{\bar z}(1/z) = \pi\delta_0$, $\Delta = 4\partial_z\partial_{\bar z}$, and (4.4), we obtain in $\mathcal{D}'(U)$

$$\partial_{\bar z} F = \Delta u - f(|z|) + c\delta_0 = \big(f(|z|) - c\delta_0\big) - f(|z|) + c\delta_0 = 0.$$

Hence F extends holomorphically across 0 and is holomorphic on all of U.

Define

$$W(z) := \pi z\, F(z).$$

Then W is holomorphic on U and $W(0) = 0$. Since $\partial_z u$ is continuous up to ∂U, so is W, and by (4.6),

$$W|_{\partial U} = \pi z \left(4\partial_z u - \frac{\Phi(|z|^2)}{z} + \frac{c}{\pi z}\right)\Big|_{\partial U} = c - \pi\Phi(|z|^2).$$

In particular, $W|_{\partial U}$ is real-valued. The imaginary part of W is therefore harmonic in U, continuous up to ∂U, and vanishes on ∂U. By the maximum principle,

$$\Im W \equiv 0 \qquad \text{in } U.$$

Since a holomorphic function with zero imaginary part is constant, there exists $\kappa \in \mathbb{R}$ such that

$$W \equiv \kappa \qquad \text{in } U.$$

Restricting to the boundary gives

$$c - \pi\Phi(|z|^2) = \kappa \qquad \text{for all } z \in \partial U.$$

Since Φ is strictly increasing, $|z|$ is constant on ∂U. Thus ∂U is contained in a circle centered at the origin. Let $R > 0$ be such that $\partial U \subset \{|z| = R\}$. Since U is open and connected, it cannot meet both $\{|z| < R\}$ and $\{|z| > R\}$, because these are disjoint open sets and $U \cap \{|z| = R\} = \varnothing$. As U is bounded, we must therefore have $U \subset \{|z| < R\}$. If this inclusion were strict, then $\{|z| < R\} \setminus U$ would contain a point, and hence, being open, would contribute a boundary point of U lying strictly inside the disk $\{|z| < R\}$, contradicting $\partial U \subset \{|z| = R\}$. Therefore

$$U = \{|z| < R\}$$

that is, U is a disk. □

Remark 4.1. The two proofs use the boundary assumptions in rather different ways.

In the first proof, once the logarithmic potential Ψ_U is written explicitly, no elliptic boundary regularity is needed to conclude that $\Psi_U \in C^1(\mathbb{R}^2 \setminus \{0\})$: this follows directly from convolution with the locally integrable kernel $\nabla \log |z|$. Thus the local radiality argument itself only uses continuity of the gradient up to the boundary. The genuine extra input appears only in the final geometric step, where one needs that, near a suitable point $x_0 \in \partial U$, the set $\partial U \cap B_r(x_0)$ contains a nontrivial boundary arc, so that the radii attained there form an interval. Accordingly, the first proof does not really need C^2 regularity; it works under any hypothesis guaranteeing such a local arc structure, for instance when U is a Jordan domain. In particular, any Lipschitz, C^1, or C^2 boundary is more than sufficient.

By contrast, the second proof does not use any boundary regularity at all. The explicit formula (4.5) already yields $u \in C^1(\mathbb{R}^2 \setminus \{0\})$, and since $u \equiv 0$ on the exterior open set $\mathbb{R}^2 \setminus \overline{U}$, continuity implies that both u and ∇u vanish on ∂U. This is exactly the boundary information needed to construct the holomorphic function W and conclude that $|z|$ is constant on ∂U. Hence the second proof recovers the theorem under the original assumptions of Abreu and Dörfler, namely for arbitrary bounded simply connected domains, with no boundary smoothness hypothesis.

5. Proof of Theorem 3

We prove that the estimate in Corollary 1.4 of [19] is sharp by constructing a family of domains which are perturbations of the disk, whose first localization eigenvalue has the same first-order behaviour as the disk, but whose distance to the class of disks is of order ε.

Proof of Theorem 3. Fix

$$\lambda_0 := 1 - e^{-\pi},$$

which is the eigenvalue of the localization operator of the unit disk D_1 associated with the ground state h_0. For ε small, define

$$P_\varepsilon(w) := 1 + \varepsilon w^2.$$

By Theorem 1, applied with $\lambda = \lambda_0$ and $P_\varepsilon(w) = 1 + \varepsilon w^2$, for all sufficiently small $|\varepsilon|$ there exists a domain U_ε such that

$$\int_{U_\varepsilon} P_\varepsilon(z) e^{\pi \bar{z} w} e^{-\pi |z|^2} \, dA(z) = \lambda_0 P_\varepsilon(w), \qquad w \in \mathbb{C}. \tag{5.1}$$

Moreover, by construction, U_ε is a C^2 perturbation of the unit disk. Hence, after possibly shrinking the range of admissible ε, its boundary can be written as

$$\partial U_\varepsilon = \{(1 + \varphi_\varepsilon(y))y : \ y \in \mathbb{S}^1\}, \tag{5.2}$$

where

$$\varphi_\varepsilon \in C^2(\mathbb{S}^1), \qquad \|\varphi_\varepsilon\|_{C^2(\mathbb{S}^1)} \lesssim |\varepsilon|.$$

$$\varepsilon \mapsto \varphi_\varepsilon$$

may in fact be chosen C^1 as a map from a neighborhood of 0 into $C^1(\mathbb{S}^1)$: this follows from the proof of Theorem 1, since both the Isakov step and the weighted quadrature-domain step are obtained by implicit-function-theorem arguments with C^1 dependence on the data, and the

polynomial

$$P_\varepsilon(z) = 1 + \varepsilon z^2$$

depends smoothly on ε. We therefore write

$$\dot{\varphi}_0(y) := \partial_\varepsilon|_{\varepsilon=0}\, \varphi_\varepsilon(y).$$

Step 1: first variation of the eigenvalue identity. We differentiate (5.1) at $\varepsilon = 0$. Since $P_\varepsilon \to 1$ in C^∞ and $\varepsilon \mapsto \varphi_\varepsilon$ is C^1 into $C^1(\mathbb{S}^1)$, all terms in (5.1) are differentiable at $\varepsilon = 0$. Since U_ε is given by a normal graph over $\mathbb{S}^1$, Reynolds' transport formula yields, for every smooth test function H,

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \int_{U_\varepsilon} H(z)\, dA(z) = \int_{\mathbb{S}^1} \dot{\varphi}_0(z) H(z)\, d\mathcal{H}^1(z).$$

Applying this to

$$H(z) = e^{\pi \bar{z} w} e^{-\pi |z|^2},$$

and differentiating the factor $P_\varepsilon(z) = 1 + \varepsilon z^2$, we obtain

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w} e^{-\pi|z|^2}\, d\mathcal{H}^1(z) + \int_{D_1} z^2 e^{\pi \bar{z} w} e^{-\pi|z|^2}\, dA(z) = \lambda_0 w^2. \tag{5.3}$$

Since D_1 is radial, the monomial z^2 is an eigenfunction of the localization operator of D_1; therefore there exists $\lambda_2 > 0$ such that

$$\int_{D_1} z^2 e^{\pi \bar{z} w} e^{-\pi|z|^2}\, dA(z) = \lambda_2 w^2.$$

Hence (5.3) becomes

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w} e^{-\pi|z|^2}\, d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \tag{5.4}$$

In fact, λ_2 can be computed explicitly. Expanding $e^{\pi \bar{z} w}$ into its power series and using orthogonality on the disk,

$$\lambda_2 = \frac{\pi^2}{2} \int_{D_1} |z|^4 e^{-\pi|z|^2}\, dA(z) = 1 - \Big(1 + \pi + \frac{\pi^2}{2}\Big) e^{-\pi}.$$

Therefore

$$\lambda_0 - \lambda_2 = \pi\Big(1 + \frac{\pi}{2}\Big) e^{-\pi} > 0,$$

so the coefficient of w^2 on the right-hand side is indeed nonzero. In particular, since $|z| = 1$ on $\mathbb{S}^1$, we may rewrite this as

$$e^{-\pi} \int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w}\, d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \tag{5.5}$$

Setting $w = 0$ gives

$$\int_{\mathbb{S}^1} \dot{\varphi}_0\, d\mathcal{H}^1 = 0. \tag{5.6}$$

Step 2: identification of the leading mode. Expanding

$$e^{\pi \bar{z} w} = \sum_{n \geq 0} \frac{\pi^n}{n!} \bar{z}^n w^n,$$

and comparing coefficients in (5.5), we find

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z)\overline{z}^n \, d\mathcal{H}^1(z) = 0 \qquad \text{for all } n \neq 2,$$

whereas the coefficient corresponding to $n = 2$ is nonzero. Since $\dot{\varphi}_0$ is real-valued, it follows that

$$\dot{\varphi}_0(e^{it}) = a\cos(2t) + b\sin(2t) \tag{5.7}$$

for some pair $(a, b) \neq (0, 0)$. In particular,

$$\int_{\mathbb{S}^1} |\dot{\varphi}_0| \, d\mathcal{H}^1 > 0. \tag{5.8}$$

Step 3: normalization of the area. The area of U_ε is given by

$$|U_\varepsilon| = \frac{1}{2}\int_{\mathbb{S}^1} (1 + \varphi_\varepsilon(y))^2 \, d\mathcal{H}^1(y). \tag{5.9}$$

Since

$$\varphi_\varepsilon = \varepsilon\dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1),$$

and (5.6) gives the vanishing of the first-order term, we obtain from (5.9)

$$|U_\varepsilon| = \pi + O(\varepsilon^2). \tag{5.10}$$

Let

$$r_\varepsilon := \sqrt{\frac{\pi}{|U_\varepsilon|}}, \qquad \widetilde{U}_\varepsilon := r_\varepsilon U_\varepsilon.$$

Then $|\widetilde{U}_\varepsilon| = \pi$, and by (5.10),

$$r_\varepsilon = 1 + O(\varepsilon^2).$$

Hence $\widetilde{U}_\varepsilon$ still admits a normal graph representation

$$\partial\widetilde{U}_\varepsilon = \{(1 + \widetilde{\varphi}_\varepsilon(y))y : \ y \in \mathbb{S}^1\},$$

with

$$\widetilde{\varphi}_\varepsilon = \varepsilon\dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1).$$

Thus the area normalization does not alter the first-order deformation.

Step 4: distance from the centered disk. Since both $\widetilde{U}_\varepsilon$ and D_1 are star-shaped with respect to the origin, the symmetric difference may be computed in polar coordinates:

$$|\widetilde{U}_\varepsilon \triangle D_1| = \frac{1}{2}\int_{\mathbb{S}^1} \left|(1 + \widetilde{\varphi}_\varepsilon(y))^2 - 1\right| \, d\mathcal{H}^1(y).$$

Using

$$(1 + \widetilde{\varphi}_\varepsilon)^2 - 1 = 2\widetilde{\varphi}_\varepsilon + \widetilde{\varphi}_\varepsilon^2 = 2\varepsilon\dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } L^\infty(\mathbb{S}^1),$$

we obtain

$$|\widetilde{U}_\varepsilon \triangle D_1| = \frac{1}{2}\int_{\mathbb{S}^1} \left|2\varepsilon\dot{\varphi}_0 + O(\varepsilon^2)\right| \, d\mathcal{H}^1.$$

Since (5.8) gives

$$\int_{\mathbb{S}^1} |\dot{\varphi}_0| \, d\mathcal{H}^1 > 0,$$

it follows that there exist constants $c, C > 0$ such that

$$c|\varepsilon| \leq |\widetilde{U}_\varepsilon \triangle D_1| \leq C|\varepsilon| \tag{5.11}$$

for all sufficiently small ε.

Step 5: comparison with arbitrary disks. Let D be any disk of area π. Then $D = D_1 + a$ for some $a \in \mathbb{C}$. We first consider disks whose centers are not too close to the origin. Since

$$|D_1 \triangle (D_1 + a)| \gtrsim \min\{|a|, 1\},$$

the triangle inequality and (5.11) imply that there exists $K \gg 1$ such that, if $|a| \geq K|\varepsilon|$, then

$$|\widetilde{U}_\varepsilon \triangle D| \geq |D_1 \triangle D| - |\widetilde{U}_\varepsilon \triangle D_1| \gtrsim |a| - C|\varepsilon| \gtrsim |\varepsilon|. \tag{5.12}$$

It remains to consider disks whose centers satisfy $|a| \leq K|\varepsilon|$. Write

$$a = \eta\varepsilon e^{i\theta}, \qquad |\eta| \leq K, \quad \theta \in [0, 2\pi).$$

We claim that, for ε sufficiently small, the translated domain

$$\widetilde{U}_\varepsilon - a$$

is still a normal graph over $\mathbb{S}^1$:

$$\partial(\widetilde{U}_\varepsilon - a) = \{(1 + \psi_\varepsilon^{\theta,\eta}(y))y : \ y \in \mathbb{S}^1\},$$

with

$$\psi_\varepsilon^{\theta,\eta} \in C^1(\mathbb{S}^1), \qquad \|\psi_\varepsilon^{\theta,\eta}\|_{C^1} \lesssim |\varepsilon|,$$

uniformly in (θ, η) with $|\eta| \leq K$. This follows because $\widetilde{U}_\varepsilon$ is already an $O(\varepsilon)$-C^1 perturbation of D_1, and the translation vector a is also of order ε. We now compute the first-order term of $\psi_\varepsilon^{\theta,\eta}$. A translation by a contributes, to first order in ε, the normal displacement

$$-\eta\Re(e^{-i\theta}y).$$

Indeed, for $y \in \mathbb{S}^1$,

$$|y - a| = \big(1 - 2\Re(\overline{y}a) + |a|^2\big)^{1/2} = 1 - \Re(\overline{y}a) + O(|a|^2),$$

uniformly in y, and here $a = \eta\varepsilon e^{i\theta}$. Therefore

$$\psi_\varepsilon^{\theta,\eta}(y) = \varepsilon\Big(\dot{\varphi}_0(y) - \eta\Re(e^{-i\theta}y)\Big) + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1), \tag{5.13}$$

uniformly for $|\eta| \leq K$.

Define

$$g^{\theta,\eta}(y) := \dot{\varphi}_0(y) - \eta\Re(e^{-i\theta}y).$$

Using (5.7), we see that $g^{\theta,\eta}$ belongs to the finite-dimensional space spanned by the first and second harmonics. More precisely,

$$g^{\theta,\eta}(e^{it}) = a\cos(2t) + b\sin(2t) - \eta\cos(t - \theta).$$

The second-harmonic part is independent of (θ,η) and nonzero, because $(a,b)\neq(0,0)$. Hence the family

$$\{g^{\theta,\eta}:\ \theta\in[0,2\pi),\ |\eta|\le K\}$$

is a compact subset of a finite-dimensional vector space which does not contain the zero function. It follows that there exists $\delta_K>0$ such that

$$\int_{\mathbb{S}^1}|g^{\theta,\eta}(y)|\,d\mathcal{H}^1(y)\ge\delta_K\qquad\text{for all }\theta\in[0,2\pi),\ |\eta|\le K. \tag{5.14}$$

Now,

$$|\widetilde{U}_\varepsilon\triangle D|=|(\widetilde{U}_\varepsilon-a)\triangle D_1|.$$

Since $\widetilde{U}_\varepsilon-a$ is star-shaped with radial graph $1+\psi_\varepsilon^{\theta,\eta}$, the same polar-coordinate computation as in Step 4 gives

$$|(\widetilde{U}_\varepsilon-a)\triangle D_1|=\frac{1}{2}\int_{\mathbb{S}^1}\left|(1+\psi_\varepsilon^{\theta,\eta}(y))^2-1\right|\,d\mathcal{H}^1(y).$$

Using (5.13),

$$(1+\psi_\varepsilon^{\theta,\eta})^2-1=2\varepsilon g^{\theta,\eta}+O(\varepsilon^2)\quad\text{in }L^\infty(\mathbb{S}^1),$$

uniformly for $|\eta|\le K$. Thus, by (5.14), there exists $c_K>0$ such that

$$|\widetilde{U}_\varepsilon\triangle D|=|(\widetilde{U}_\varepsilon-a)\triangle D_1|\ge c_K|\varepsilon| \tag{5.15}$$

whenever $|a|\le K|\varepsilon|$. Combining (5.12) and (5.15), we conclude that

$$\inf_{\substack{D\text{ disk}\\|D|=\pi}}|\widetilde{U}_\varepsilon\triangle D|\gtrsim|\varepsilon|.$$

Since $r_\varepsilon=1+O(\varepsilon^2)$, the dilation from U_ε to $\widetilde{U}_\varepsilon$ changes the set only by

$$|\widetilde{U}_\varepsilon\triangle U_\varepsilon|=O(\varepsilon^2),$$

and therefore does not affect the first-order distance estimate. Thus the area-normalized family $\widetilde{U}_\varepsilon$ remains at symmetric-difference distance $\gtrsim|\varepsilon|$ from the class of disks, while its spectral data differ from those of the disk only at the same perturbative scale as before the normalization. This shows that the estimate in Corollary 1.4 of [19] is sharp.

□

6. Local maximizers for the Faber–Krahn problem in the Gaussian STFT setting

We work throughout in the Bargmann–Fock representation. Thus, if $F\in\mathcal{F}^2(\mathbb{C})$ is the Bargmann transform of $f\in L^2(\mathbb{R})$, then

$$|\mathcal{V}_\varphi f(z)|^2=|F(z)|^2e^{-\pi|z|^2},$$

up to the normalization already fixed in the previous sections. Accordingly, for $F\in\mathcal{F}^2(\mathbb{C})$ we set

$$u_F(z):=|F(z)|^2e^{-\pi|z|^2}.$$

For a measurable set $\Omega \subset \mathbb{C}$, let

$$\lambda_{1,\varphi}(\Omega) := \sup_{\|f\|_2=1} \int_\Omega |\mathcal{V}_\varphi f(z)|^2 \, dz.$$

We say that Ω is a *local maximizer* for $\lambda_{1,\varphi}$ under the measure constraint if there exists $\delta_0 > 0$ such that, whenever $E \subset \mathbb{C}$ is measurable, $|E| = |\Omega|$, and

$$\int_{\mathbb{C}} |\mathbf{1}_\Omega - \mathbf{1}_E| \, e^{-\pi|z|^2} \, dz < \delta_0,$$

one has

$$\lambda_{1,\varphi}(\Omega) \geq \lambda_{1,\varphi}(E).$$

We now prove the local rigidity statement announced in the introduction.

Proof of the local-maximizer theorem. Let $\Sigma \subset \mathbb{C}$ be a local maximizer for $\lambda_{1,\varphi}$ under the measure constraint, and let $F \in \mathcal{F}^2(\mathbb{C})$ be a normalized first eigenfunction for the localization operator associated with Σ. Equivalently,

$$\int_\Sigma H(z)\overline{F(z)} \, d\mu(z) = \lambda_{1,\varphi}(\Sigma) \int_{\mathbb{C}} H(z)\overline{F(z)} \, d\mu(z)$$

for every $H \in \mathcal{F}^2(\mathbb{C})$, where

$$d\mu(z) := e^{-\pi|z|^2} \, dA(z).$$

Step 1: Σ is a superlevel set of u_F.

We claim that there exists $t_0 \geq 0$ such that, up to sets of measure zero,

$$\Sigma = \{u_F > t_0\}.$$

Indeed, suppose this fails. Then there exists t_0 such that

$$A := \{u_F > t_0\} \qquad \text{satisfies} \qquad |A| = |\Sigma|,$$

but A and Σ do not agree a.e. Since the weighted symmetric-difference distance is absolutely continuous with respect to Lebesgue measure on bounded sets, we may choose measurable sets

$$S \subset A \setminus \Sigma, \qquad T \subset \Sigma \setminus A, \qquad |S| = |T| > 0,$$

with

$$\int_{\mathbb{C}} |\mathbf{1}_S + \mathbf{1}_T| e^{-\pi|z|^2} \, dz < \delta_0.$$

Define

$$\Sigma' := (\Sigma \setminus T) \cup S.$$

Then $|\Sigma'| = |\Sigma|$ and

$$\int_{\mathbb{C}} |\mathbf{1}_\Sigma - \mathbf{1}_{\Sigma'}| e^{-\pi|z|^2} \, dz < \delta_0.$$

On the other hand, by construction,

$$u_F(z) > t_0 \quad \text{on } S, \qquad u_F(z) \leq t_0 \quad \text{on } T,$$

hence

$$\int_{\Sigma'} u_F\,dA = \int_{\Sigma} u_F\,dA + \int_S u_F\,dA - \int_T u_F\,dA > \int_{\Sigma} u_F\,dA.$$

Since F is admissible for the variational problem defining $\lambda_{1,\varphi}(\Sigma')$, this implies

$$\lambda_{1,\varphi}(\Sigma') \geq \int_{\Sigma'} u_F\,dA > \int_{\Sigma} u_F\,dA = \lambda_{1,\varphi}(\Sigma),$$

contradicting the local maximality of Σ. This proves the claim.

In particular, u_F is constant on $\partial\Sigma$, in the sense that there exists $c_\Sigma > 0$ with

$$u_F(z) = c_\Sigma \qquad \text{for } z \in \partial\Sigma. \tag{6.1}$$

Step 2: simplicity of the first eigenvalue.

We claim that the first eigenvalue $\lambda_{1,\varphi}(\Sigma)$ is simple.

Suppose instead that $G \in \mathcal{F}^2(\mathbb{C})$ is another first eigenfunction, linearly independent of F. By the previous step, Σ is also a superlevel set of u_G, hence there exists $c'_\Sigma > 0$ such that

$$u_G(z) = c'_\Sigma \qquad \text{for } z \in \partial\Sigma.$$

Equivalently,

$$|F(z)|^2 e^{-\pi|z|^2} = c_\Sigma, \qquad |G(z)|^2 e^{-\pi|z|^2} = c'_\Sigma, \qquad z \in \partial\Sigma.$$

Dividing, we obtain

$$\frac{|F(z)|^2}{|G(z)|^2} = \frac{c_\Sigma}{c'_\Sigma} \qquad \text{for } z \in \partial\Sigma.$$

Now we use the standard fact that first eigenfunctions may be chosen without zeros in Σ: by Step 1, Σ is a superlevel set of u_F, hence $u_F > 0$ in the interior of Σ. Since $u_F = |F|^2 e^{-\pi|z|^2}$ and the Gaussian factor never vanishes, it follows that F has no zeros in Σ. The same argument applies to G. Therefore F/G is holomorphic and nonvanishing in Σ, and the function

$$\log\left|\frac{F}{G}\right| = \frac{1}{2}\log\frac{|F|^2}{|G|^2} = \frac{1}{2}\log\frac{u_F}{u_G}$$

is harmonic in Σ. Since the Gaussian factors cancel, this function is constant on $\partial\Sigma$. By the maximum principle, it is constant on each connected component of Σ. Hence

$$\left|\frac{F}{G}\right| \equiv \text{constant} \qquad \text{on each connected component of } \Sigma.$$

Since F/G is holomorphic and has constant modulus on each connected component, it is constant there. Thus

$$F = \gamma_j G \qquad \text{on each connected component of } \Sigma.$$

By analyticity, the same relation holds on all of $\mathbb{C}$. By unique continuation, F and G are linearly dependent, a contradiction. This proves simplicity.

Step 3: a first-variation identity.

Since Σ is a level set of the smooth function u_F, its boundary is Lipschitz (indeed C^1 away from the critical set of u_F, which is enough for the divergence argument below). Let ψ be

holomorphic on an open neighborhood V of $\overline{\Sigma}$, and write

$$\psi = u + iv$$

with u, v real-valued. Consider the vector field

$$X_\psi := (u, -v).$$

Since ψ is holomorphic, the Cauchy–Riemann equations give

$$u_x = v_y, \qquad u_y = -v_x,$$

hence

$$\operatorname{div} X_\psi = u_x - v_y = 0 \qquad \text{in } V.$$

Because u_F is constant on $\partial\Sigma$, we obtain

$$0 = \int_{\partial\Sigma} u_F\, X_\psi \cdot \nu \, d\mathcal{H}^1 = \int_\Sigma \operatorname{div}(u_F X_\psi)\, dA = \int_\Sigma \langle \nabla u_F, X_\psi \rangle \, dA.$$

Since $u_F = |F|^2 e^{-\pi|z|^2}$, this becomes

$$0 = \int_\Sigma \left(\langle \nabla |F|^2, X_\psi \rangle - 2\pi |F|^2 \langle z, X_\psi \rangle \right) d\mu. \tag{6.2}$$

Let

$$A(z) := F'(z)\overline{F(z)}.$$

Since

$$\partial_x |F|^2 = A + \overline{A} = 2\Re A, \qquad \partial_y |F|^2 = iA - i\overline{A} = -2\Im A,$$

a direct computation gives

$$\langle \nabla |F|^2, X_\psi \rangle = 2\,\Re\big(\psi\, F' \overline{F}\big).$$

Likewise, writing $z = x + iy$, we have

$$\langle z, X_\psi \rangle = \Re(z\psi).$$

Substituting these expressions into (6.2), we obtain

$$0 = \Re \int_\Sigma \psi(z) \Big(F'(z)\overline{F(z)} - \pi z\, |F(z)|^2 \Big)\, d\mu(z).$$

Applying the same identity with $i\psi$ in place of ψ, we also obtain

$$0 = \Im \int_\Sigma \psi(z) \Big(F'(z)\overline{F(z)} - \pi z\, |F(z)|^2 \Big)\, d\mu(z).$$

Hence

$$0 = \int_\Sigma \psi(z) \Big(F'(z)\overline{F(z)} - \pi z\, |F(z)|^2 \Big)\, d\mu(z). \tag{6.3}$$

Taking complex conjugates, we may equivalently write

$$0 = \int_\Sigma \psi(z)\, F(z) \Big(\overline{F'(z)} - \pi z\, \overline{F(z)} \Big)\, d\mu(z)$$

for every holomorphic ψ on a neighborhood of $\overline{\Sigma}$.

Step 4: F' is again a first eigenfunction.

We now show that F' is an eigenfunction associated with the same first eigenvalue $\lambda_{1,\varphi}(\Sigma)$. For every $w \in \mathbb{C}$, set

$$K_w(z) := e^{\pi \overline{w} z}.$$

By Step 1, after replacing Σ by the corresponding superlevel-set representative if necessary, we may suppose that

$$\Sigma = \{u_F > c_\Sigma\}.$$

In particular, $u_F \geq c_\Sigma > 0$ on $\overline{\Sigma}$. Since

$$u_F(z) = |F(z)|^2 e^{-\pi|z|^2},$$

it follows that F has no zeros on $\overline{\Sigma}$. By continuity and the compactness of $\overline{\Sigma}$, there exists an open neighborhood $V \supset \overline{\Sigma}$ on which F has no zeros. Hence, for every $w \in \mathbb{C}$, the function

$$\psi_w(z) := \frac{K_w(z)}{F(z)}$$

is holomorphic on V. Applying the identity from Step 3 with $\psi = \psi_w$, we get

$$0 = \int_\Sigma e^{\pi \overline{w} z} \Big(\overline{F'(z)} - \pi z \, \overline{F(z)} \Big) \, d\mu(z).$$

Hence

$$\int_\Sigma e^{\pi \overline{w} z} \overline{F'(z)} \, d\mu(z) = \int_\Sigma \pi z \, e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z). \tag{6.4}$$

On the other hand, from the eigenvalue equation for F, taking

$$G_0(z) := z e^{\pi \overline{w} z},$$

we obtain

$$\int_\Sigma z e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z) = \lambda_{1,\varphi}(\Sigma) \int_{\mathbb{C}} z e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z).$$

Using the reproducing property of the Fock space,

$$\int_{\mathbb{C}} e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z) = \overline{F(w)},$$

and differentiating with respect to $\overline{w}$, we get

$$\pi \int_{\mathbb{C}} z e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z) = \overline{F'(w)}.$$

Therefore

$$\int_\Sigma \pi z e^{\pi \overline{w} z} \overline{F(z)} \, d\mu(z) = \lambda_{1,\varphi}(\Sigma) \, \overline{F'(w)}.$$

Combining this with (6.4), we conclude that

$$\int_\Sigma e^{\pi \overline{w} z} \overline{F'(z)} \, d\mu(z) = \lambda_{1,\varphi}(\Sigma) \, \overline{F'(w)}, \qquad w \in \mathbb{C}.$$

This is exactly the eigenvalue equation for F'. Thus F' is a first eigenfunction as claimed.

By simplicity of the first eigenvalue, proved in Step 2, it follows that

$$F' = \theta F$$

for some constant $\theta \in \mathbb{C}$. Solving this ODE gives

$$F(z) = Ce^{\theta z},$$

for some nonzero constant C.

Step 5: reduction to the constant eigenfunction.

Up to translation of the set Σ, we may absorb the exponential factor $e^{\theta z}$ and reduce to the case

$$F \equiv 1.$$

Indeed, in the Bargmann–Fock model, translating the underlying domain corresponds to composing the eigenfunction with the appropriate Weyl factor; thus, after a translation, we may normalize away the linear exponential term.

Therefore, after translation, the ground state h_0 is an eigenfunction of the localization operator of Σ. Since Σ is a level set of $u_F(z) = e^{-\pi|z|^2}$, it follows at once that Σ is either an annulus or a disk.

Step 6: exclusion of annuli and conclusion.

It remains to show that among radial sets of prescribed measure, the only local maximizer is the disk. If Σ is radial, write

$$\widetilde{\Sigma} := \{r > 0 : \ re^{it} \in \Sigma \text{ for some } t\}.$$

Since Σ has Lipschitz boundary, $\widetilde{\Sigma}$ is a finite union of disjoint intervals:

$$\widetilde{\Sigma} = \bigcup_j (a_j, b_j).$$

Then

$$\int_\Sigma e^{-\pi|z|^2}\, dz = 2\pi \int_{\widetilde{\Sigma}} re^{-\pi r^2}\, dr = \sum_j 2\pi \int_{a_j}^{b_j} re^{-\pi r^2}\, dr = \sum_j \left(e^{-\pi a_j^2} - e^{-\pi b_j^2}\right).$$

Under the constraint

$$|\Sigma| = 2\pi \int_{\widetilde{\Sigma}} r\, dr = \pi \sum_j (b_j^2 - a_j^2)$$

fixed, this quantity is maximized by placing all the radial mass as close as possible to the origin, namely by taking

$$\widetilde{\Sigma} = (0, R)$$

for the unique $R > 0$ such that $|D_R| = |\Sigma|$. Therefore the only radial maximizer is the centered disk.

We conclude that every local maximizer of $\lambda_{1,\varphi}$ under the measure constraint is, up to translation, a disk. □

7. Global maximizers for the Faber–Krahn problem in the Gaussian STFT setting

In this section we prove existence of global maximizers for the Gaussian time-frequency concentration problem under a measure constraint, and then combine that existence result with the

local rigidity theorem from the previous section to recover the disk as the unique maximizer, up to translation.

Since the Bargmann transform is unitary, the variational problem may be written as

$$M(s) := \sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega) = \sup_{\|F\|_{\mathcal{F}^2}=1} I_F(s).$$

Lemma 7.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k\} \subset \mathbb{C}$ satisfy $|a_k| \to \infty$. Then*

$$T_{a_k} G \rightharpoonup 0 \qquad \text{weakly in } \mathcal{F}^2(\mathbb{C}),$$

and in particular

$$\langle F, T_{a_k} G\rangle_{\mathcal{F}^2} \to 0.$$

Consequently,

$$\|F + T_{a_k} G\|_{\mathcal{F}^2}^2 = \|F\|_{\mathcal{F}^2}^2 + \|G\|_{\mathcal{F}^2}^2 + o(1).$$

More generally, if $|a_k - b_k| \to \infty$, then

$$\langle T_{a_k} F, T_{b_k} G\rangle_{\mathcal{F}^2} \to 0.$$

Proof. Using unitarity,

$$\langle F, T_{a_k} G\rangle = \langle T_{-a_k} F, G\rangle.$$

Now $T_{-a_k} F$ is bounded in $\mathcal{F}^2$, and (**??**) shows that its associated densities drift to infinity. Equivalently, for every compact $K \subset \mathbb{C}$,

$$\sup_{z\in K} |T_{-a_k} F(z)| e^{-\pi|z|^2/2} \to 0.$$

Hence $T_{-a_k} F \to 0$ locally uniformly, and therefore weakly in $\mathcal{F}^2$. This proves the first claim. The norm decoupling follows by expanding the square. The general case reduces to the first by writing

$$\langle T_{a_k} F, T_{b_k} G\rangle = \langle F, T_{b_k - a_k} G\rangle.$$

□

Lemma 7.2. *Fix $J \geq 1$, let $G^{(1)}, \dots, G^{(J)} \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k^{(j)}\}_{k\geq 1} \subset \mathbb{C}$ for $1 \leq j \leq J$ satisfy*

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad \text{whenever } j \neq \ell.$$

For each k, set

$$S_k := \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)}.$$

Then the following hold.

(i) For every $R > 0$ and every $1 \leq j \leq J$,

$$\sup_{z\in D_R(a_k^{(j)})} \left| u_{S_k}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z) \right| \to 0 \qquad (k \to \infty).$$

(ii) For every $R > 0$, if

$$E_{k,R} := \mathbb{C} \setminus \bigcup_{j=1}^{J} D_R(a_k^{(j)}),$$

then

$$\limsup_{k\to\infty} \sup_{z\in E_{k,R}} u_{S_k}(z) \le \left(\sum_{j=1}^{J} \sup_{\zeta\in\mathbb{C}\setminus D_R} u_{G^{(j)}}(\zeta)^{1/2} \right)^2 .$$

In particular, given $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that for every $R \ge R_\varepsilon$,

$$\sup_{z\in E_{k,R}} u_{S_k}(z) \le \varepsilon$$

for all sufficiently large k.

Proof. We repeatedly use the identity

$$u_{T_a F}(z) = u_F(z-a), \qquad z, a \in \mathbb{C},$$

proved earlier.

Proof of (i). Fix $R > 0$ and $j \in \{1, \dots, J\}$. Write

$$S_k = T_{a_k^{(j)}} G^{(j)} + H_{k,j}, \qquad H_{k,j} := \sum_{\ell\neq j} T_{a_k^{(\ell)}} G^{(\ell)}.$$

Then for every $z \in \mathbb{C}$,

$$\begin{aligned} \left| u_{S_k}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z) \right| &= e^{-\pi|z|^2} \left| |T_{a_k^{(j)}} G^{(j)}(z) + H_{k,j}(z)|^2 - |T_{a_k^{(j)}} G^{(j)}(z)|^2 \right| \\ &\le 2\, u_{T_{a_k^{(j)}} G^{(j)}}(z)^{1/2} u_{H_{k,j}}(z)^{1/2} + u_{H_{k,j}}(z). \end{aligned}$$

By Lemma 2.1(ii),

$$u_{T_{a_k^{(j)}} G^{(j)}}(z) \le \|G^{(j)}\|^2_{\mathcal{F}^2} \qquad \text{for all } z, k.$$

Thus it is enough to show that

$$\sup_{z\in D_R(a_k^{(j)})} u_{H_{k,j}}(z) \to 0.$$

For $z \in D_R(a_k^{(j)})$ and $\ell \neq j$, we have

$$z - a_k^{(\ell)} \in D_R(a_k^{(j)} - a_k^{(\ell)}).$$

Since $|a_k^{(j)} - a_k^{(\ell)}| \to \infty$, it follows that

$$\operatorname{dist}\left(D_R(a_k^{(j)} - a_k^{(\ell)}), 0 \right) \to \infty.$$

Because $u_{G^{(\ell)}} \in C_0(\mathbb{C})$, we obtain

$$\sup_{z\in D_R(a_k^{(j)})} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z) = \sup_{\zeta\in D_R(a_k^{(j)} - a_k^{(\ell)})} u_{G^{(\ell)}}(\zeta) \to 0.$$

Now

$$u_{H_{k,j}}(z)^{1/2} \le \sum_{\ell\neq j} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z)^{1/2},$$

so the preceding convergence implies

$$\sup_{z\in D_R(a_k^{(j)})} u_{H_{k,j}}(z)^{1/2} \to 0.$$

This proves (i).

Proof of (ii). Fix $R > 0$ and $z \in E_{k,R}$. Then $z \notin D_R(a_k^{(j)})$ for every j, hence $|z - a_k^{(j)}| \geq R$. Therefore

$$u_{T_{a_k^{(j)}} G^{(j)}}(z) = u_{G^{(j)}}(z - a_k^{(j)}) \leq \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta).$$

Using again the elementary inequality

$$u_{\sum_{j=1}^J F_j}(z)^{1/2} \leq \sum_{j=1}^J u_{F_j}(z)^{1/2},$$

we get

$$u_{S_k}(z)^{1/2} \leq \sum_{j=1}^J u_{T_{a_k^{(j)}} G^{(j)}}(z)^{1/2} \leq \sum_{j=1}^J \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta)^{1/2}.$$

Squaring yields

$$u_{S_k}(z) \leq \left(\sum_{j=1}^J \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta)^{1/2} \right)^2.$$

Taking the supremum over $z \in E_{k,R}$ proves the displayed estimate in (ii). Since each $u_{G^{(j)}} \in C_0(\mathbb{C})$, the right-hand side tends to 0 as $R \to \infty$. The final claim follows. □

Lemma 7.3. *Let $\{S_k\} \subset \mathcal{F}^2(\mathbb{C})$ be bounded and let $\{R_k\} \subset \mathcal{F}^2(\mathbb{C})$ satisfy*

$$\|u_{R_k}\|_{L^\infty(\mathbb{C})} \to 0.$$

Then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \to 0.$$

More precisely, if $\sup_k \|S_k\|_{\mathcal{F}^2} \leq M$, then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \leq 2M \, \|u_{R_k}\|_{L^\infty(\mathbb{C})}^{1/2} + \|u_{R_k}\|_{L^\infty(\mathbb{C})}.$$

Proof. For every $z \in \mathbb{C}$,

$$\begin{aligned} \left| u_{S_k+R_k}(z) - u_{S_k}(z) \right| &= e^{-\pi|z|^2} \left| |S_k(z) + R_k(z)|^2 - |S_k(z)|^2 \right| \\ &\leq 2\, u_{S_k}(z)^{1/2} u_{R_k}(z)^{1/2} + u_{R_k}(z). \end{aligned}$$

By Lemma 2.1(ii),

$$u_{S_k}(z) \leq \|S_k\|_{\mathcal{F}^2}^2 \leq M^2.$$

Hence

$$\left| u_{S_k+R_k}(z) - u_{S_k}(z) \right| \leq 2M \, \|u_{R_k}\|_{L^\infty}^{1/2} + \|u_{R_k}\|_{L^\infty}.$$

Taking the supremum in z gives the claim. □

Lemma 7.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then the function*

$$s \mapsto I_F(s)$$

is finite, nondecreasing, and continuous on $[0, \infty)$.

Proof. By Lemma 2.5, $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$, so $I_F(s)$ is finite for every $s \geq 0$. Monotonicity is immediate from the definition.

To prove continuity, let $u := u_F$, and for $t \geq 0$ define

$$\mu_u(t) := \big|\{z \in \mathbb{C} : \ u(z) > t\}\big|.$$

Since $u \in C_0(\mathbb{C})$, the bathtub principle shows that for each $s \geq 0$,

$$I_F(s) = \int_0^s u^*(\tau)\, d\tau,$$

where u^* is the decreasing rearrangement of u. In particular, $u^* \in L^1_{\mathrm{loc}}([0,\infty))$, and therefore I_F is continuous on $[0,\infty)$. □

The next result is the profile decomposition adapted to the present setting. Since it is the main compactness input of the section, we record it explicitly.

Proposition 7.1 (Profile decomposition in the Fock space)**.** *Let $\{F_k\}_{k\geq 1} \subset \mathcal{F}^2(\mathbb{C})$ be a bounded sequence. Then, after passing to a subsequence, there exist profiles*

$$G^{(j)} \in \mathcal{F}^2(\mathbb{C}), \qquad j \geq 1,$$

and sequences of points

$$a_k^{(j)} \in \mathbb{C}, \qquad j \geq 1,\ k \geq 1,$$

such that:

(i) for each $j \neq \ell$,

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad (k \to \infty);$$

(ii) for every $J \geq 1$, if

$$R_k^{(J)} := F_k - \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)},$$

then

$$T_{-a_k^{(j)}} R_k^{(J)} \to 0 \qquad \textit{locally uniformly on } \mathbb{C}$$

for every $1 \leq j \leq J$;

(iii) for every $J \geq 1$,

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1);$$

(iv)

$$\lim_{J\to\infty} \left(\limsup_{k\to\infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} \right) = 0.$$

Proof. We argue by the standard iterative extraction scheme, but we spell out the points used later.

If

$$\limsup_{k\to\infty} \|u_{F_k}\|_{L^\infty(\mathbb{C})} = 0,$$

there is nothing to prove: take all profiles equal to zero. Otherwise choose $a_k^{(1)} \in \mathbb{C}$ such that

$$u_{F_k}(a_k^{(1)}) \geq \frac{1}{2} \|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

By (**??**), replacing F_k by $T_{-a_k^{(1)}}F_k$ if necessary, we may suppose $a_k^{(1)} = 0$. Since $\{F_k\}$ is bounded in $\mathcal{F}^2$, the reproducing-kernel bound implies that $\{F_k\}$ is a normal family. Passing to a subsequence, $F_k \to G^{(1)}$ locally uniformly, with $G^{(1)} \neq 0$ by the choice of the centering. Set

$$R_k^{(1)} := F_k - G^{(1)}.$$

If

$$\limsup_{k\to\infty} \|u_{R_k^{(1)}}\|_{L^\infty(\mathbb{C})} = 0,$$

we stop. Otherwise choose $a_k^{(2)}$ so that

$$u_{R_k^{(1)}}(a_k^{(2)}) \geq \frac{1}{2}\|u_{R_k^{(1)}}\|_{L^\infty(\mathbb{C})}.$$

The local uniform convergence $R_k^{(1)} \to 0$ on compact sets forces $|a_k^{(2)}| \to \infty$. Then $T_{-a_k^{(2)}}R_k^{(1)}$ is again bounded in $\mathcal{F}^2$, so after passing to a subsequence it converges locally uniformly to some nonzero profile $G^{(2)}$. We set

$$R_k^{(2)} := R_k^{(1)} - T_{a_k^{(2)}}G^{(2)}.$$

Proceeding inductively, we obtain profiles $G^{(j)}$, translations $a_k^{(j)}$, and remainders $R_k^{(J)}$. Let us verify the properties in the statement.

Proof of (i). Suppose $j < \ell$. At the ℓ-th step, the sequence

$$T_{-a_k^{(\ell)}}R_k^{(\ell-1)}$$

converges locally uniformly to the nonzero profile $G^{(\ell)}$. On the other hand, for every $m < \ell$, property (ii) already proved up to stage $\ell - 1$ gives

$$T_{-a_k^{(m)}}R_k^{(\ell-1)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}.$$

If $|a_k^{(\ell)} - a_k^{(m)}|$ failed to tend to infinity along a subsequence, then after passing to a further subsequence the difference $a_k^{(\ell)} - a_k^{(m)}$ would stay bounded, and therefore local uniform convergence of $T_{-a_k^{(m)}}R_k^{(\ell-1)}$ to 0 would force $T_{-a_k^{(\ell)}}R_k^{(\ell-1)} \to 0$ locally uniformly as well, a contradiction. Thus

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad \text{for } j \neq \ell.$$

Proof of (ii). Fix $J \geq 1$ and $1 \leq j \leq J$. By construction, at the j-th extraction step we have

$$T_{-a_k^{(j)}}R_k^{(j-1)} \to G^{(j)} \qquad \text{locally uniformly on } \mathbb{C}.$$

Subtracting the j-th extracted profile yields

$$T_{-a_k^{(j)}}R_k^{(j)} = T_{-a_k^{(j)}}R_k^{(j-1)} - G^{(j)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}.$$

Now let $J > j$. Since

$$R_k^{(J)} = R_k^{(j)} - \sum_{\ell=j+1}^{J} T_{a_k^{(\ell)}}G^{(\ell)},$$

we get

$$T_{-a_k^{(j)}} R_k^{(J)} = T_{-a_k^{(j)}} R_k^{(j)} - \sum_{\ell=j+1}^{J} T_{a_k^{(\ell)}-a_k^{(j)}} G^{(\ell)}.$$

The first term tends to 0 locally uniformly by the previous paragraph. For each fixed $\ell > j$, property (i) and the identity

$$u_{T_b G}(z) = u_G(z-b)$$

imply that $T_{a_k^{(\ell)}-a_k^{(j)}} G^{(\ell)} \to 0$ locally uniformly, since $|a_k^{(\ell)} - a_k^{(j)}| \to \infty$. Thus

$$T_{-a_k^{(j)}} R_k^{(J)} \to 0 \qquad \text{locally uniformly on } \mathbb{C},$$

which is exactly (ii).

Proof of (iii). For fixed J, write

$$F_k = \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)} + R_k^{(J)}.$$

Expanding the $\mathcal{F}^2$-norm squared and using Lemma 7.1, together with the pairwise divergence of the translation parameters from (i), we obtain

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1).$$

Proof of (iv). Suppose, on the contrary, that (iv) fails. Then there exist $\delta > 0$, a sequence $J_m \to \infty$, and indices $k_m \to \infty$ such that

$$\|u_{R_{k_m}^{(J_m)}}\|_{L^\infty(\mathbb{C})} \geq \delta \qquad \text{for every } m.$$

Choose $b_m \in \mathbb{C}$ so that

$$u_{R_{k_m}^{(J_m)}}(b_m) \geq \delta/2.$$

We claim that for every fixed $j \geq 1$,

$$|b_m - a_{k_m}^{(j)}| \to \infty \qquad (m \to \infty).$$

Indeed, if this failed for some fixed j, then after passing to a subsequence we would have $|b_m - a_{k_m}^{(j)}| \leq C$. Since $J_m \geq j$ for all large m, property (ii) gives

$$T_{-a_{k_m}^{(j)}} R_{k_m}^{(J_m)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}.$$

Evaluating at $z = b_m - a_{k_m}^{(j)}$, which remains in a fixed compact set, we would obtain

$$u_{R_{k_m}^{(J_m)}}(b_m) \to 0,$$

contradicting $u_{R_{k_m}^{(J_m)}}(b_m) \geq \delta/2$. Thus the claim holds.

Now the sequence

$$T_{-b_m} R_{k_m}^{(J_m)}$$

is bounded in $\mathcal{F}^2(\mathbb{C})$, so after passing to a subsequence it converges locally uniformly to some $H \in \mathcal{F}^2(\mathbb{C})$. By construction,

$$u_H(0) = \lim_{m\to\infty} u_{T_{-b_m} R_{k_m}^{(J_m)}}(0) = \lim_{m\to\infty} u_{R_{k_m}^{(J_m)}}(b_m) \geq \delta/2,$$

so $H \neq 0$.

Moreover, for every fixed j,

$$|b_m - a_{k_m}^{(j)}| \to \infty,$$

so the new centers b_m are asymptotically disjoint from all previously extracted profiles. Therefore H could be extracted as one more nontrivial profile, contradicting the maximality of the decomposition procedure. This proves (iv). □

We now turn to the global problem.

Theorem 6. *For every $s > 0$, there exists $F_* \in \mathcal{F}^2(\mathbb{C})$ with $\|F_*\|_{\mathcal{F}^2} = 1$ such that*

$$I_{F_*}(s) = M(s).$$

Equivalently, there exists a measurable set $\Omega_* \subset \mathbb{C}$ with $|\Omega_*| = s$ such that

$$\lambda_{1,\varphi}(\Omega_*) = M(s).$$

Proof. Let $\{F_k\} \subset \mathcal{F}^2(\mathbb{C})$ be a maximizing sequence:

$$\|F_k\|_{\mathcal{F}^2} = 1, \qquad I_{F_k}(s) \to M(s).$$

By replacing F_k with a Weyl translate if necessary, and using the translation invariance (**??**), we may suppose that

$$u_{F_k}(0) = \|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

Applying Proposition 7.1, after passing to a subsequence we may write

$$F_k = \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)} + R_k^{(J)}$$

for every fixed J, with the conclusions of that proposition.

Fix $J \geq 1$ and let $\Omega_k^{(J)}$ be a measurable set of measure s such that

$$I_{F_k}(s) = \int_{\Omega_k^{(J)}} u_{F_k}\, dA.$$

By the bathtub principle, we may choose $\Omega_k^{(J)}$ to be a superlevel set of u_{F_k}.

Let $\varepsilon > 0$. Choose $R > 0$ so large that

$$\int_{\mathbb{C}\setminus D_R} u_{G^{(j)}}\, dA < \varepsilon \qquad \text{for all } 1 \leq j \leq J.$$

Since the translation parameters are pairwise divergent, for k large enough the disks

$$D_R(a_k^{(1)}), \ldots, D_R(a_k^{(J)})$$

are pairwise disjoint. Write

$$B_{k,j} := D_R(a_k^{(j)}), \qquad E_{k,j} := \Omega_k^{(J)} \cap B_{k,j}, \qquad E_{k,0} := \Omega_k^{(J)} \setminus \bigcup_{j=1}^{J} B_{k,j}.$$

Set

$$s_{k,j} := |E_{k,j}|, \qquad 0 \le j \le J.$$

Then

$$\sum_{j=0}^{J} s_{k,j} = s.$$

Fix

$$S_k^{(J)} := \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)}.$$

By Proposition 7.1(iii), the sequence $\{S_k^{(J)}\}_k$ is bounded in $\mathcal{F}^2(\mathbb{C})$. By Proposition 7.1(iv), if J is taken sufficiently large then

$$\limsup_{k\to\infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon^2.$$

Lemma 7.3 therefore gives

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} \to 0 \qquad (k \to \infty),$$

and in particular, after increasing k if necessary,

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon. \tag{7.1}$$

Now let

$$E_{k,0} = \Omega_k^{(J)} \setminus \bigcup_{j=1}^{J} B_{k,j} \subset \mathbb{C} \setminus \bigcup_{j=1}^{J} D_R(a_k^{(j)}).$$

Lemma 7.2(i) gives, for each fixed $j \in \{1, \dots, J\}$,

$$\sup_{z\in B_{k,j}} \left|u_{S_k^{(J)}}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z)\right| \to 0 \qquad (k \to \infty),$$

while Lemma 7.2(ii) yields, after taking R sufficiently large and then k sufficiently large,

$$\sup_{z\in E_{k,0}} u_{S_k^{(J)}}(z) \le \varepsilon.$$

Combining these bounds with (7.1), we conclude that for all sufficiently large k,

(a) on each $B_{k,j}$,

$$u_{F_k}(z) \le u_{T_{a_k^{(j)}} G^{(j)}}(z) + 2\varepsilon;$$

(b) on $E_{k,0}$,

$$u_{F_k}(z) \le 2\varepsilon.$$

Therefore

$$I_{F_k}(s) = \int_{\Omega_k^{(J)}} u_{F_k}\, dA \le \sum_{j=1}^{J} \int_{E_{k,j}} u_{T_{a_k^{(j)}} G^{(j)}}\, dA + +2\varepsilon \sum_{j=0}^{J} |E_{k,j}|$$

$$= \sum_{j=1}^{J} \int_{E_{k,j}-a_k^{(j)}} u_{G^{(j)}}\, dA + +2\varepsilon s \le \sum_{j=1}^{J} I_{G^{(j)}}(s_{k,j}) + 2\varepsilon s.$$

Passing to a further subsequence if necessary, we may suppose

$$s_{k,j} \to \sigma_j \in [0,s] \qquad (k \to \infty)$$

for each $1 \le j \le J$. By continuity of $I_{G^{(j)}}(\cdot)$ in the measure parameter, given by Lemma 7.4, we obtain

$$M(s) \le \sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j) + 2\varepsilon s.$$

Since $\varepsilon > 0$ is arbitrary,

$$M(s) \le \sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j). \tag{7.2}$$

Now use Lemma 2.5. For each j,

$$I_{G^{(j)}}(\sigma_j) = \|G^{(j)}\|_{\mathcal{F}^2}^2 I_{\widehat{G}^{(j)}}(\sigma_j), \qquad \widehat{G}^{(j)} := \frac{G^{(j)}}{\|G^{(j)}\|_{\mathcal{F}^2}}$$

if $G^{(j)} \neq 0$, and

$$I_{\widehat{G}^{(j)}}(\sigma_j) \le M(s)$$

because $\sigma_j \le s$ and $I_{\widehat{G}^{(j)}}(\cdot)$ is nondecreasing. Hence

$$I_{G^{(j)}}(\sigma_j) \le \|G^{(j)}\|_{\mathcal{F}^2}^2\, M(s).$$

Summing and using Proposition 7.1(iii),

$$\sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j) \le M(s) \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 \le M(s).$$

Together with (7.2), this gives

$$M(s) \le \sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j) \le M(s)$$

for every J. Therefore equality holds throughout, and in particular

$$\sum_{j\ge 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = 1.$$

Moreover, for every j with $G^{(j)} \neq 0$,

$$I_{\widehat{G}^{(j)}}(s) \ge I_{\widehat{G}^{(j)}}(\sigma_j) = M(s).$$

Since $M(s)$ is the supremum over all unit-norm functions, it follows that

$$I_{\widehat{G}^{(j)}}(s) = M(s).$$

Thus any nonzero normalized profile is a maximizer. Taking one such profile and denoting it by F_*, we obtain

$$\|F_*\|_{\mathcal{F}^2} = 1, \qquad I_{F_*}(s) = M(s).$$

Finally, let Ω_* be a superlevel set of u_{F_*} with $|\Omega_*| = s$ and

$$\int_{\Omega_*} u_{F_*}\, dA = I_{F_*}(s) = M(s).$$

Then

$$\lambda_{1,\varphi}(\Omega_*) \geq \int_{\Omega_*} u_{F_*}\, dA = M(s),$$

while by definition $\lambda_{1,\varphi}(\Omega_*) \leq M(s)$. Hence

$$\lambda_{1,\varphi}(\Omega_*) = M(s),$$

and Ω_* is a global maximizer. □

We may now identify the global maximizers.

Theorem 7. *For every $s > 0$, every global maximizer for*

$$\sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega)$$

is, up to translation, a disk.

Proof. Let Ω_* be a global maximizer of measure s. Then Ω_* is automatically a local maximizer under the same measure constraint. Therefore the rigidity theorem proved in the previous section applies: every local maximizer is, up to translation, a disk.

Hence every global maximizer is a disk up to translation. Since the problem is translation invariant, this completely characterizes the set of global maximizers. □

As an immediate consequence, we recover the classical Nicola–Tilli theorem.

Corollary 7.1. *Among all measurable sets of prescribed measure, the disk maximizes Gaussian time-frequency concentration:*

$$\lambda_{1,\varphi}(\Omega) \leq \lambda_{1,\varphi}(D)$$

whenever $|\Omega| = |D|$ and D is a disk.

Proof. Existence of a global maximizer is given by Theorem 6, and Theorem 7 shows that every such maximizer is a disk up to translation. Since translation preserves both measure and the value of $\lambda_{1,\varphi}$, the claim follows. □

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